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Beyond the Needle: Attention-Based Multimodal Deep Learning for Anemia Screening Through Ablation Studies and Comparative Benchmarking

By

Jasmin Sultana Shimu (0112230353)

Sayma Sultana (0112230948)

Md Fazley Rabbi (0112230944)

Tasmim Jahan Rimvy (0112230846)

Supervisor: Dr. Jannatun Noor Mukta

Co-Supervisor: Khushnur Binte Jahangir

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Abstract

Anemia affects a large part of the world's population, yet its standard diagnosis requires a blood sample and laboratory support that are often unavailable in remote or resource-limited settings. This thesis investigates whether smartphone recordings of the fingertip, the palm and the fingernail can support non-invasive anemia screening, and how these sources should be combined. Using a public dataset of 909 young adults, ImageNet-pretrained ResNet-18 models were first fine-tuned on each modality separately with patient-level stratified splits. For the video modalities, an ablation then varied the number of sampled frames (8, 16 and 32) and the training strategy (class-imbalance weighting, data augmentation, or both). The best single-modality models reached test accuracies of 73.87% for fingertip videos, 67.27% for nail photographs and 67.20% for palm videos. A tri-modal fusion framework was then studied through a component-wise ablation in which per-modality ResNet-18 encoders were combined by concatenation or self-attention fusion, frame features were aggregated by mean pooling or a bidirectional LSTM, and demographic and symptom variables were optionally added. The ablation was run on a common three-class test set of 123 participants under single-split and five-fold protocols. No component had a consistent effect: six fairly comparable configurations reached a mean five-fold macro-F1 between 0.380 and 0.459, close to the 0.407 of a logistic regression on demographic and symptom variables alone, and no difference was statistically supported. The image-only model with BiLSTM pooling and self-attention fusion had the highest macro-F1 (0.459 ± 0.075; fold-averaged macro AUC-ROC 0.756) and is reported as a provisional candidate, but it identified only one of the seven participants with moderate anemia. One configuration with much higher scores was found unreliable and excluded. Benchmarked against eleven re-implemented published methods on the same three-class task, the selected model was not exceeded on macro-F1, but seven methods lay within one standard deviation of it. The conclusions are further limited by single-site data, small test sets and an inconsistency found in the label file used for the binary experiments. The findings indicate that the three recordings contain a detectable signal for anemia, but the performance is not yet adequate for stand-alone clinical screening.

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Chapter 1

Introduction

Anemia is a reduction in the concentration of hemoglobin, the protein that carries oxygen in red blood cells, below the level required to meet physiological needs. It is among the most widespread nutritional and hematological conditions worldwide and, according to the World Health Organization, affected an estimated 30.7% of women aged 15–49 years in 2023 and 39.8% of children aged 6–59 months in 2019[2]. This thesis investigates whether short smartphone recordings of three superficial body sites, namely the fingertip, the palm and the fingernail, can support non-invasive anemia screening when they are analyzed with deep learning, and how the information from these sites should be combined. This chapter introduces the problem, identifies the gap that motivates the work, states the objectives and research questions, and outlines how the remainder of the thesis addresses them.

1.1 Project Overview

Background and Context

The reference method for diagnosing anemia is the measurement of hemoglobin concentration in a blood sample. The procedure is accurate, but it is invasive, requires trained personnel and laboratory or point-of-care equipment, and causes discomfort that discourages repeated testing[3]. These requirements are difficult to meet in remote or resource-limited settings, where the burden of anemia is often greatest and where screening is therefore least available[4, 5]. A screening method that needs only a smartphone camera would remove most of these obstacles. Its purpose would not be to replace laboratory diagnosis, but to identify people who should be referred for a confirmatory test.

The clinical basis for such a method is long established: reduced hemoglobin lowers the red coloration of well-perfused tissue, so pallor of the nail bed, the palm and the conjunctiva is a recognized sign of anemia[6, 7]. Computer-vision studies have exploited this sign by estimating hemoglobin or classifying anemia from photographs of the conjunctiva, fingernails and palms[5, 7, 8], and smartphone applications have estimated hemoglobin from fingertip video recorded under controlled illumination[9]. Deep convolutional neural networks have extended this line of work because they learn discriminative features

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directly from images instead of relying on hand-designed color statistics[6, 7, 10].

Problem Statement

Despite this progress, building a dependable image-based anemia screening model remains difficult, for three reasons that shape the design of this thesis.

First, the visual signal is subtle and is easily confounded by illumination, skin tone, camera settings and the position of the body part. A model trained on one body site must therefore separate a small color change from large irrelevant variation.

Second, the available labeled data are limited. Medical image datasets with reference hemoglobin measurements are small compared with general-purpose image collections, and the anemic participants are usually a minority, so models are prone to overfitting and to favoring the majority class.

Third, a single body site provides only a partial view. Each site is affected by different sources of noise, and a video of the fingertip contains temporal information (the changing color as blood flows under pressure and release) that a single photograph does not. It is not obvious how much each site contributes, how many video frames are needed, or how the sites should be combined.

The central problem addressed in this thesis can therefore be stated as follows: *how can fingertip videos, palm videos and fingernail photographs be processed and combined with a deep learning model to detect anemia, and what is the individual and joint value of each source of information?*

1.2 Motivation

The literature reviewed in Chapter 2 shows that most image-based anemia studies analyze a single anatomical site, most commonly the conjunctiva, the fingernail or the palm[11, 12, 13, 14]. Studies that combine several sources of information typically pair one image type with electronic health records or laboratory-derived variables[15], or fuse still photographs of several body sites[6]. Work that treats video, and in particular the temporal dimension of a fingertip recording, is comparatively scarce[16]. The public dataset of Valles-Coral et al.[1] is unusual in providing, for the same 909 young adults, a fingertip video, a palm video and a nail photograph together with hemoglobin values, age, sex and symptoms. Published analyses of this dataset that we are aware of concentrate on the nail photographs alone[13], so the contribution of the two video modalities, the effect of temporal sampling and aggregation, and the benefit of combining all three sites remain insufficiently characterized. These unanswered questions define the gap that this thesis addresses.

The motivation for studying fusion is both practical and methodological. Practically, a screening workflow that records three short clips is inexpensive, and combining sites may reduce the dependence on any single, possibly poorly captured, recording. Methodologi-

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cally, convolutional networks pretrained on large image collections can be transferred to small medical datasets[17], and there are several established ways to merge the resulting representations, from simple concatenation to attention between modalities[18, 19, 20]. Comparing these options under controlled conditions is a sound way to learn which design choices matter, and it motivates the experimental structure adopted here: first to establish what each modality can achieve alone, then to examine the factors that influence the video modalities, and finally to test fusion.

1.3 Objectives

The aim of this thesis is to develop and evaluate a deep learning framework that detects anemia from fingertip videos, palm videos and fingernail photographs, and to determine which design choices improve its performance. This aim is pursued through the following objectives.

1. To prepare the public multimodal dataset for learning, including verification of the available files, construction of binary and three-class labels, and patient-level stratified splitting into training, validation and test sets.
2. To establish single-modality baselines in which an ImageNet-pretrained ResNet-18 is fine-tuned separately on each modality.
3. To examine, for the video modalities, the effect of the number of sampled frames and of two training strategies, class-imbalance weighting and data augmentation.
4. To design and compare multimodal fusion models that combine the three modalities, contrasting concatenation with self-attention fusion and mean pooling with bidirectional LSTM aggregation of frame features, with and without demographic and symptom variables.
5. To evaluate all models with standard classification metrics, to compare them with tabular-only baselines, and to state clearly what the evidence does and does not support.
6. To benchmark the selected model against alternative model frameworks under an identical cohort, label set, data split and evaluation protocol, so that the proposed design can be judged against established alternatives.

Objectives 1 to 5 correspond to four research questions, and objective 6 to a fifth, each of which is answered in Chapter 4. Objective 6 concerns the comparative benchmarking, whose protocol is described in Section 3.2.8 and whose results are reported in Section 4.3.5.

RQ1 How well can each single modality (fingernail photograph, palm video, fingertip video) discriminate anemic from non-anemic participants with a fine-tuned ResNet-18?

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RQ2 How do the number of sampled frames and the training strategy (class-imbalance weighting, augmentation) affect the video-based models?

RQ3 How do concatenation and self-attention fusion of the three modalities compare, and how does the choice of temporal aggregation (mean pooling or bidirectional LSTM) influence the result?

RQ4 What do demographic and symptom variables contribute, alone and together with the images?

RQ5 How does the selected model compare with published methods for anemia detection that were re-implemented on the same three-class task, and where do their errors differ?

1.4 Methodology

The study uses one public dataset, collected at a single site with a single smartphone model from participants aged 18 to 25. All models share an ImageNet-pretrained ResNet-18 as the visual encoder. The work is organized in two stages. In the first, single-modality binary classifiers (Normal versus Anemic, the latter combining the mild and moderate categories) are trained and refined through an ablation of frame count and training strategy. In the second, the three modalities are fused in a tri-modal network, and a component-wise ablation of the fusion module, the temporal pooling and the demographic and symptom variables is run in a three-class setting (Normal, Mild, Moderate) under single-split and five-fold protocols, with a binary experiment as a reference. The best-supported configuration is then analysed as the selected model. In a third stage, it is benchmarked against eleven published methods that were re-implemented on the same three-class task. Explainability methods, clinical deployment and external validation are outside the scope of the thesis and are discussed as future work.

1.5 Project Outcome

The thesis makes the following contributions, each of which is supported by the experiments in Chapter 4.

- A documented preprocessing and patient-level evaluation pipeline for the three modalities of the dataset, including modality-specific frame selection and nail-bed cropping.
- Single-modality ResNet-18 baselines and an ablation showing how frame count and training strategy affect the palm and fingertip models.
- A tri-modal fusion framework with two interchangeable fusion modules, two temporal aggregation methods and an optional demographic and symptom branch, together

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with a component-wise ablation study evaluated against tabular-only baselines and a documented selection of a provisional best model.

- A comparative benchmarking of the selected model against eleven re-implemented published methods, with class-wise errors and a screening view.
- A critical account of the limitations of the evidence, including the small moderate-anemia class, the instability of single-split results, an unreliable experiment that was identified and excluded, and an inconsistency identified in the label file used for the binary experiments.

1.6 Organization of the Report

The remainder of this report follows the progression from problem to evidence. Chapter 2 presents the medical and technical preliminaries, reviews similar applications and related research, and states the gap that this work addresses. Chapter 3 translates the objectives into a design: the requirements, the detailed methodology (design alternatives, dataset and labels, preprocessing, baselines, ablation studies, the fusion framework and the benchmarking protocol), the project plan and the task allocation. Chapter 4 describes the environment setup and the testing and evaluation protocol, and reports and discusses the results. Chapter 5 documents the standards, design constraints, cost analysis and complex engineering problem mapping required by the degree programme. Chapter 6 summarizes the work, states its limitations and outlines future work.

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Chapter 2

Background

Chapter 1 identified the need for a non-invasive anemia screening method and the open questions around combining several body sites. This chapter provides the knowledge required to address those questions. It first summarizes how anemia is defined and recognized clinically and introduces the deep learning concepts used in the thesis (preliminaries). It then reviews similar applications and related research, and closes with a gap analysis that establishes what is already known, what is unresolved, and how that motivates the design in Chapter 3.

2.1 Preliminaries

This section introduces the clinical and technical concepts on which the rest of the report depends.

2.1.1 Anemia and Its Clinical Assessment

Anemia is a condition in which the number of circulating red blood cells, or their hemoglobin content, is insufficient to meet the oxygen demands of the body. It is diagnosed by comparing the hemoglobin concentration with reference thresholds that depend on age, sex and pregnancy status. The World Health Organization defines anemia, for example, as a hemoglobin concentration below 13.0 g/dL in men aged 15 years and above, below 12.0 g/dL in non-pregnant women of the same age and below 11.0 g/dL in pregnant women and in children aged 6 to 59 months[21]. Severity is graded by further thresholds, and moderate anemia, with hemoglobin in the range of approximately 8.0 to 10.9 g/dL in adults, can already cause functional impairment[21]. Typical symptoms are fatigue, dizziness, headache, shortness of breath and cold hands and feet, and visible pallor of the skin and of the tissue under the fingernails becomes more evident as severity increases.

Two aspects of the labels used in this thesis should be noted at the outset. The dataset described in Section 3.2.2 grades participants as Normal, Mild (*Leve*) or Moderate (*Moderada*) using thresholds that were defined with a regional blood bank to suit the

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local population, including its altitude, rather than by directly applying the general WHO table[1]. In addition, the reference hemoglobin values were obtained with a non-invasive pulse co-oximetry device and not with a laboratory blood count. The labels therefore represent a clinical reference standard of good but not laboratory-grade quality, which is a limitation shared by every model trained on them.

2.1.2 Convolutional Networks and Transfer Learning

Convolutional neural networks (CNNs) learn hierarchies of spatial features from images through stacked convolutional layers. ResNet-18 is a compact member of the residual network family; it contains 18 weighted layers, batch normalization and ReLU activations, and uses skip connections that ease the optimization of deep networks[22]. Because medical datasets are small, networks are rarely trained from scratch. In transfer learning, a network pretrained on a large natural-image collection such as ImageNet is adapted to the new task, either by training only a new classification layer on frozen features or by fine-tuning some or all layers[17]. This thesis fine-tunes an ImageNet-pretrained ResNet-18 for every visual modality, which keeps the model small and suitable for the limited computing resources described in Chapter 5.

2.1.3 Temporal Aggregation of Video Frames

A video is processed as a sequence of frames. A frame-level encoder, here ResNet-18, converts each frame into a feature vector, and a temporal aggregation step combines the sequence into one video-level representation. The simplest option is mean pooling, which averages the frame features and ignores their order. A bidirectional long short-term memory network (BiLSTM) reads the sequence in both directions and can in principle model order-dependent changes[23]. The choice matters here because the fingertip signal arises from color change over time, whereas the palm video mainly shows a hand that opens.

2.1.4 Multimodal Fusion

Multimodal learning combines information from several input types. Fusion strategies are commonly classified by the stage at which information is merged: early fusion combines raw inputs, intermediate (feature-level) fusion combines learned representations, and late fusion combines the decisions of separately trained models[18, 19, 20]. This thesis uses intermediate fusion. Two fusion modules are compared: concatenation, which joins the modality embeddings into one vector and leaves interactions to the classifier, and self-attention fusion, in which the embeddings are treated as tokens of a transformer layer so that each modality can weight the others[24].

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2.1.5 Tabular Models for Clinical Variables

Demographic and symptom variables are structured data. Logistic regression and random forests are standard baselines for such data, and a small multilayer perceptron can map the same variables into an embedding that is fused with the image features. Comparing a model that uses only these variables with the image-based models indicates how much of the predictive signal the images provide.

2.2 Literature Review

The studies below are grouped by the kind of evidence they use, because the relevant question for this thesis is how far each kind of evidence can take non-invasive anemia detection.

2.2.1 Similar Applications

Early smartphone approaches estimated hemoglobin from physical signals. HemaApp estimates hemoglobin from fingertip video recorded with the phone camera under LED illumination, using chromatic analysis and without machine learning, and reported a sensitivity of 85.7% for anemia screening[9]. More recent work has reported real-world implementation of a smartphone application augmented with artificial intelligence, together with personalization for hemoglobin self-monitoring[25]. These studies establish that video of the fingertip carries hemoglobin-related information, but they depend on controlled capture conditions and use hand-crafted signal processing rather than learned visual features.

2.2.2 Related Research

The research literature on image-based anemia detection is organized below by the body site and the kind of data used.

Image-based detection from a single body site. A large body of work classifies anemia from still images of one site. For the conjunctiva, studies have applied color-space conversion, region-of-interest segmentation and classical or deep classifiers to detect anemia, including in children[26, 27]. For the fingernail, Das and Bhattacharjee analyzed 286 nail images using intensity, histogram and white-spot features and noted dependence on image quality and the absence of clinical data[11]. For the palm, Appiahene et al. compared several classifiers on 527 palm images in the CIELAB color space and reported very high accuracy with a CNN and with k-nearest neighbors, while recommending that fingertip or nail information be added in future work[28]. A mobile health solution combining a CNN and a support vector machine on nail bed, conjunctiva and palm images from 567 subjects reported accuracy above 90% for mild anemia but used images only[29]. These studies show that each site is informative. They also differ in dataset size, capture protocol and evaluation design, so their accuracy values are not directly comparable with

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one another or with those obtained in this thesis, and very high accuracies reported on small datasets should be interpreted with caution.

Work on video and on the present dataset. Fingertip video has been studied as a modality in its own right, for example in a dataset for non-invasive anemia diagnosis with a ResNet-18 classifier[16]. The dataset used in this thesis, released by Valles-Coral et al., provides fingertip videos, palm videos and nail photographs for 909 young adults from Peru together with hemoglobin values, age, sex and symptoms[1]. The same group has evaluated deep networks on the nail photographs of this population and reported a test accuracy of about 0.70 and an area under the ROC curve of about 0.74 with a DenseNet architecture[13]. This result is the closest published reference for the nail experiments reported here, with the caveat that the data partition may differ. The dataset paper provides only the raw recordings, so frame extraction and cropping must be designed by the user.

Multimodal and clinical-data approaches. Fusion has been explored mainly between images and structured data. Ramzan et al. fuse conjunctiva image features with electronic health record features, selected with a random forest, and report improved performance over unimodal inputs[15]. Studies that use only clinical or demographic variables, such as predictions of childhood anemia from survey data, of anemia in geriatric patients, or from complete blood count parameters, reach high accuracy on their own populations[30, 31, 32, 33, 34, 35], but they depend on laboratory results or survey variables and therefore do not provide non-invasive screening. Closest to the present work, Lin et al. combined conjunctiva, palm and fingernail photographs in an attention-based fusion network that also uses a body-part label, and reported an accuracy of 0.849 and an F1-score of 0.828 on a retrospective dataset, falling to 0.716 and 0.788 on a prospective sample of 101 patients[6]. This result shows that fusing several body sites is feasible, and it also shows how much performance can drop outside the development data. It uses still photographs and not video. Reviews of deep-learning-based information fusion describe the early, intermediate and late strategies used in this thesis[18, 19, 20].

Earlier signal-based and small-data anemia models. Before image-based deep learning became common, non-invasive hemoglobin estimation relied on dedicated optical equipment. Ding et al. built a spectrophotometer with nine LEDs and recorded fingertip spectra from 109 volunteers; principal component analysis reduced the spectra and a back-propagation neural network predicted hemoglobin with a correlation coefficient of 0.94[36]. Such systems need custom hardware, which a smartphone-only screening method avoids. Other studies use small tabular or image-derived data. One feed-forward network was trained on the sex, hemoglobin and red, green and blue pixel percentages of only 104 cases and reported 97% test accuracy[37], and another combined a Kaggle table of blood indices with palm images, reaching 99.58% accuracy with an AlexNet-based network that

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uses channel and spatial attention[38]. Accuracy values of this size on datasets of this size should be read with caution, as noted above for single-site studies.

Anemia studies that combine images with numerical data. Several recent studies fuse images with numerical variables. Arul et al. combined nail and eye images with blood parameters (MCV, hemoglobin, MCH and sex) in a two-branch network with a DenseNet169 image branch, and reported accuracy of 94 to 95%[39]. Yagnika et al. proposed a three-stage framework for clinical, image and genetic data in which generative networks (CTGAN and StyleGAN2) augment the data from 1,000 to 2,000 samples before attention-based fusion, and reported 99.2% accuracy[40]. These approaches depend on laboratory-derived or genetic inputs, or on synthetic samples, and none of them uses video.

Fusion strategies in deep learning. A large literature compares where fusion should take place. Systematic reviews of imaging combined with electronic health records describe early, joint and late fusion and give implementation guidelines[41], and a short overview of fusion for medical images describes the steps of information extraction, representation learning and fusion[42]. On human activity recognition, Gadzicki et al. compared early and late fusion of RGB video, optical flow and skeleton data with an Inception I3D network and found that early fusion gave the best results[43]; this shows that the best fusion stage depends on the task, which is one reason to test it empirically here. More recent medical fusion models use attention and pretrained foundation models. MedFMIT fuses a medical image with its text description using adapter-enhanced transformer encoders on frozen BiomedCLIP features[44], HyPCA-Net combines residual adaptive attention with a cascaded dual-view attention module and was evaluated on ten public benchmarks[45], and an explainable framework for early diagnosis fuses images, clinical text and ECG signals through attention[46]. These studies fuse different data types from those used in this thesis, but they support the choice of attention-based fusion as a modern alternative to concatenation.

Medical image-to-image fusion. A separate line of work fuses images from different scanners, such as CT, MRI, PET and SPECT, into a single image, rather than fusing features for classification. Several reviews cover the imaging modalities, databases, fusion categories and quality metrics[47, 48, 49], and one compares generative approaches (GANs, VAEs and diffusion models)[50]. Classical methods fuse CT and MRI using transform-domain decomposition, for example the non-subsampled contourlet transform with correlation-coefficient and structural-similarity rules[51], the stationary wavelet transform with a Wiener filter[52], or wavelet fusion after multi-resolution rigid registration[53]; a comparison of SWT-, NSST-, NSCT- and CNN-based fusion found the CNN-based approach to perform best[54]. Learning-based methods followed, including a hybrid layer decomposition with CNN-based feature mapping[55] and transformer-based

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unsupervised networks such as MACTFusion and MATR, both evaluated on the Harvard whole-brain atlas[56, 57]. Related reviews and applications consider multi-modality fusion for segmentation[58] and a multi-phase CT model for liver cancer[59]. This family produces fused images for clinicians and does not address classification of anemia, so it is relevant to this thesis mainly as background on how complementary views can be combined.

Attention and feature fusion in other tasks. Attention-based fusion has been applied well beyond medicine. Zhou et al. replaced concatenation and summation in an encoder-decoder with attentional feature fusion for surgical instrument segmentation[60], and Lao et al. used co-attention and a multistep cross-modal fusion network for visual question answering[61]. For chest X-rays, a dual-stream network combines a convolutional branch with a bidirectional LSTM branch over image patches and reached 93.73% accuracy[62]. The last study is of interest because it also uses a bidirectional recurrent layer, although on patches of a single image rather than on frames of a video.

Explainability. Because clinical use requires that predictions can be inspected, surveys of explainable artificial intelligence in healthcare categorize the available techniques for medical imaging and text[63], and several of the studies above add Grad-CAM, SHAP or attention maps to their models[46, 62].

Blood-smear microscopy (laboratory background). Anemia has also been studied with laboratory microscopy, where individual red blood cells are classified. Deep networks have been used to classify red blood cells for sickle cell disease with accuracy above 99%[64], shape descriptors with an ensemble of classifiers have been used to identify anemic diseases in smear images[65], and a neural network on shape variables has been used to detect sickle and elliptocyte cells[66]. These methods require stained smears and a microscope, so they belong to laboratory diagnosis and are not comparable with the screening setting of this thesis.

2.3 Gap Analysis

The literature supports three observations. First, each of the nail, palm and fingertip sites is individually informative, and deep networks are a natural tool for analyzing them. Second, most published models use one site, or one image type plus clinical records, and the few studies that fuse several body sites use still photographs, so the value of combining video-based sites recorded with the same device on an open dataset is not well quantified. Third, the temporal information in video is exploited by signal-processing methods but is less often studied with learned frame-level features, and little is known about how many frames are needed or whether sequence modeling outperforms simple averaging.

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This thesis addresses these points on a dataset that provides all three visual modalities per participant. It contributes a systematic comparison of frame sampling and training strategies for the video modalities, a fusion framework and component-wise ablation that compare concatenation with self-attention, mean pooling with a BiLSTM and the effect of demographic and symptom variables, and the selection of a provisional best model. It does not claim to be the first work to combine visual sites, and its conclusions are limited by the single-site, single-device dataset.

2.4 Summary

Anemia is defined by hemoglobin thresholds, and the labels used here follow locally defined thresholds applied to non-invasive measurements. Deep networks pretrained with transfer learning, temporal aggregation of frame features and intermediate fusion provide the technical toolkit. The literature shows that single sites and signal-based methods are informative but leaves open how fingertip video, palm video and nail photographs should be combined. Chapter 3 therefore designs experiments that first measure each modality, then study the factors affecting the video modalities, and finally compare fusion strategies.

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Chapter 3

Project Design

Chapter 2 concluded that the contribution of each visual site, the role of temporal sampling and the benefit of fusion are not yet well characterized for fingertip video, palm video and nail photographs. This chapter describes the design that was adopted to answer the research questions of Section 1.3. Requirements and context are set out first (Section 3.1). The detailed methodology and design follows (Section 3.2): the design alternatives and the reasons for the choices made, the dataset and labels, the preprocessing, the single-modality baselines that answer RQ1, the ablation of frame sampling and training strategy that answers RQ2, the fusion framework and its ablation that answer RQ3 and RQ4, and the protocol of the comparative benchmarking. The project plan and the task allocation close the chapter (Sections 3.3 and 3.4). The testing and evaluation protocol that is common to all experiments is defined in Section 4.2 of Chapter 4.

3.1 Requirement Analysis

Requirement analysis identifies what a screening system built around the model should do. It is included here because it determines the inputs, outputs and quality attributes that the experimental design must respect, even though only the model component is implemented and evaluated in this thesis.

3.1.1 Functional and Non-Functional Requirements

The functional requirements are as follows.

FR1: Multimodal input. The system accepts a fingertip video, a palm video, a nail image and, optionally, patient information (age, sex and symptoms).

FR2: Data preparation. The system extracts frames from the videos, resizes and normalizes the images, and isolates the relevant region where required.

FR3: Feature extraction. A convolutional neural network extracts visual features, and a multilayer perceptron encodes the patient information.

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FR4: Fusion. The extracted features are combined into one representation.

FR5: Classification. The system performs binary (non-anemic or anemic) or three-class classification and is trained with a cross-entropy loss.

FR6: Output. The system reports the predicted class and its probability.

The non-functional requirements set design targets for the screening system: an accuracy of at least 80%, a prediction time of about five seconds after data submission, restricted access and data security, scalability for mass screening, and an interface that medical staff can use without training. These targets are used in Chapter 4 as reference points; the prediction time and the interface were not evaluated in this thesis.

3.1.2 Context Diagram

Figure 3.1 shows the intended context of a deployed screening platform as a single process that exchanges information with four external parties.

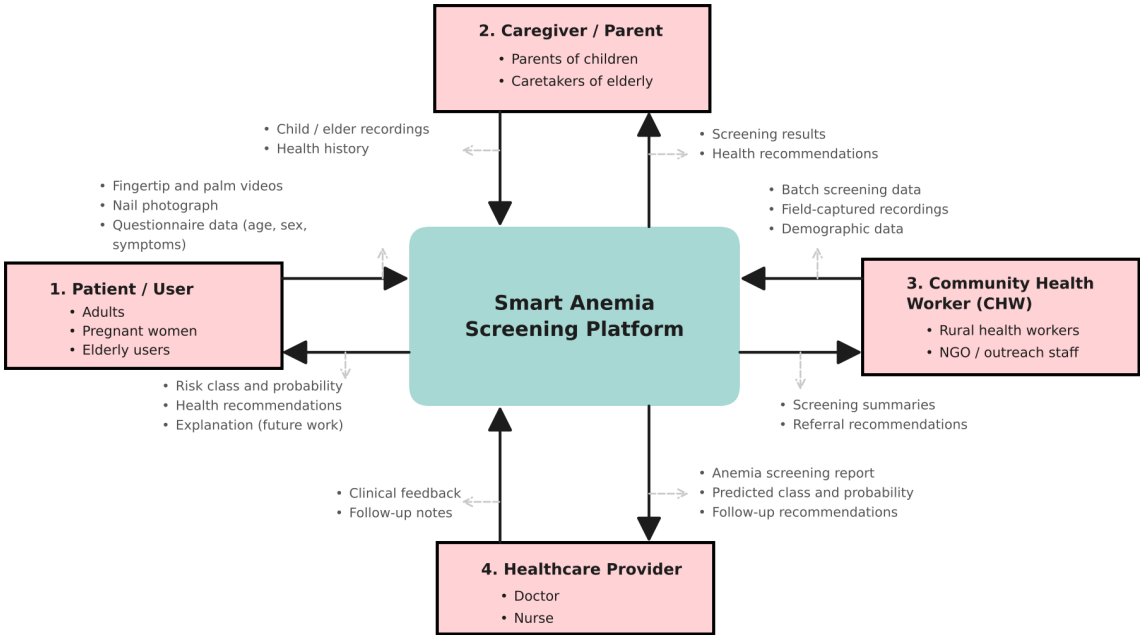


Figure 3.1: Context diagram of the intended anemia screening platform.

Figure 3.1 shows the intended context of a deployed screening platform as a single process that exchanges information with four external parties. Patients and users supply a fingertip video, a palm video, a nail photograph and questionnaire data (age, sex and symptoms), and receive the predicted risk class with its probability and health recommendations. Caregivers and parents act for children or older people, supplying recordings and health history and receiving the screening results and recommendations. Community health workers upload batch screening data, field-captured recordings and demographic data, and receive screening summaries and referral recommendations. Healthcare providers receive the anemia screening report, the predicted class and probability, and follow-up recommendations,

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and return clinical feedback and follow-up notes. The diagram describes the long-term application context. The dataset available for this thesis contains adults aged 18 to 25 only, and no explanation module was implemented, so the experiments evaluate the model component for this population and do not address the other user groups shown.

3.1.3 Level-1 Data Flow Diagram

Figure 3.2 presents the data flow of the baseline pipeline used for the single-modality experiments.

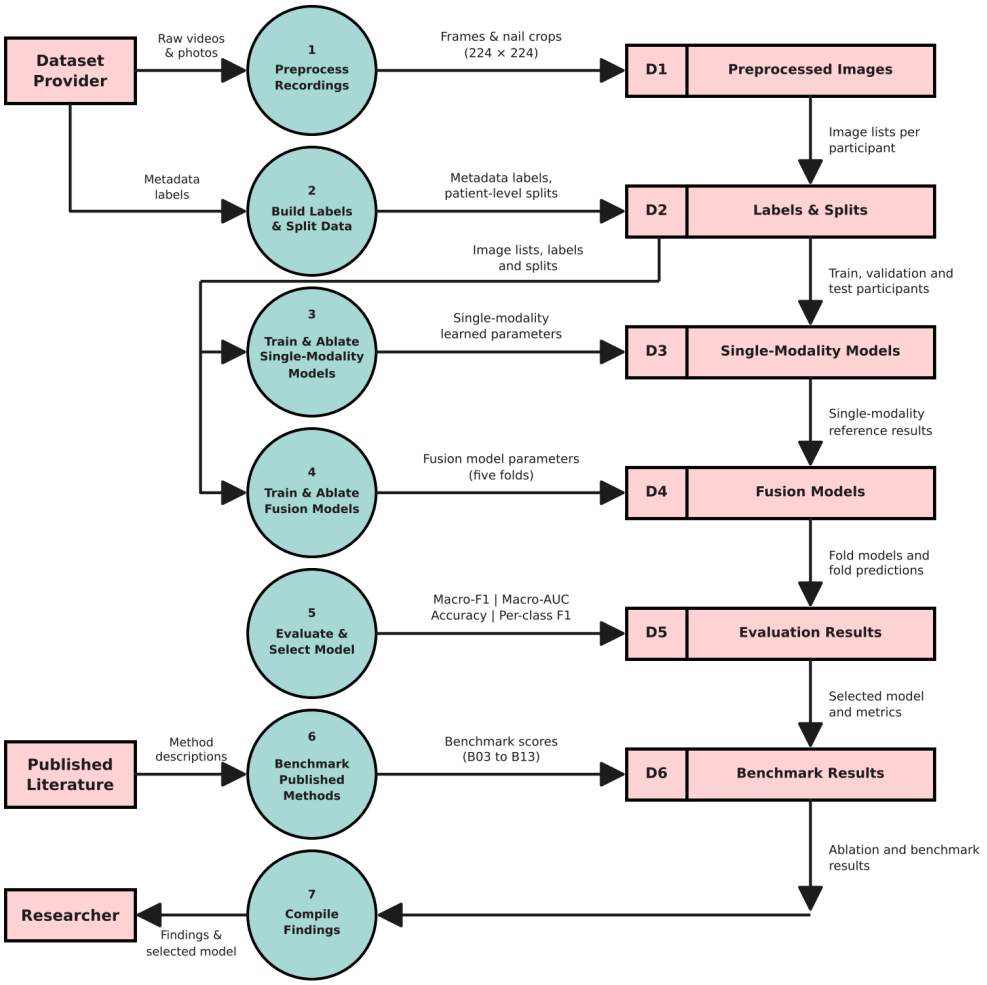


Figure 3.2: Level-1 data flow diagram of the single-modality baseline pipeline.

Figure 3.2 presents the data flow of the experimental pipeline used in this thesis. The dataset provider supplies the raw recordings to Process 1, which extracts 32 fingertip frames and 16 palm frames per participant, localizes and crops the nail photograph, and resizes every image to 224 × 224 pixels; the results are stored in D1. The participant metadata go to Process 2, which builds the binary and three-class labels and creates the patient-level stratified splits and the five folds; these are stored with the image lists in D2. Process 3 fine-tunes the single-modality ResNet-18 models and runs the ablation of

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frame sampling and training strategy, and stores the models in D3. Process 4 trains the tri-modal fusion models and runs the ablation of fusion module, temporal pooling and demographic features under the single-split and five-fold protocols, and stores the fold models in D4. Process 5 evaluates the models on the fixed test set, compares them with the tabular baselines and selects the provisional model (D5). Process 6 re-implements the eleven published methods from their descriptions in the literature and scores them against the selected model (D6). Process 7 compiles the findings, the selected model and the limitations of the evidence for the researchers.

3.2 Detailed Methodology and Design

This section describes the methodology in detail. It begins with the main design decisions and the alternatives that were available for each, and then defines the data, the preprocessing, the models and the experiments in the order in which they were carried out.

3.2.1 Design Alternatives and Rationale

Every design decision of the study had alternatives. Table 3.1 lists the main decisions, the options that were available, the option that was chosen and the reason. Where the choice was settled by experiment, the table points to the ablation that compared the options; where it was settled by the resources or the data, the reason is stated directly.

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Table 3.1: Main design decisions, the alternatives and the rationale for the choice.

Decision	Alternatives	Choice	Rationale
Visual encoder	Training a CNN from scratch; fine-tuning a pretrained ResNet-18; heavier pretrained backbones	ImageNet-pretrained ResNet-18, fine-tuned	Small medical datasets rarely support training from scratch (Section 2.1). ResNet-18 is compact and fits free-tier GPUs (Section 5.3). Other model frameworks are compared in the benchmarking stage (Section 3.2.8).
Frames of a video	Independent frames; mean pooling; BiLSTM	Frame encoder with mean pooling or BiLSTM, both evaluated	The fingertip signal arises from colour change over time, so order may matter, while averaging is the simplest baseline. Settled by the fusion ablation (Section 4.3.3).
Fusion level	Early (raw inputs); intermediate (features); late (decisions)	Intermediate fusion	The inputs differ in form (32 frames, 16 frames, one image), which makes raw-input fusion impractical. Late fusion combines separately trained decisions and cannot learn interaction between modalities.
Fusion module	Concatenation; self-attention	Both compared	Concatenation leaves interaction to the classifier, whereas self-attention learns it explicitly. Settled by the fusion ablation.
Patient information	Images only; images with twelve demographic and symptom features	Both compared	Tests whether the non-image features add information (RQ4).
Label space	Binary (Normal or Anemic); three-class (Normal, Mild, Moderate)	Both used	The binary label matches the screening decision, and the three-class label preserves severity.
Evaluation protocol	Single split; five-fold cross-validation	Both used	A single split is cheap but noisy on a seven-participant minority class, so five-fold results carry the conclusions.
Computing	Local GPU workstation; paid cloud GPUs; free-tier cloud GPUs	Free-tier Colab and Kaggle	No cost and easy reproduction by others, at the price of limited tuning and repeated runs (Section 5.3).

The decisions that were not settled by experiment are the choice of ResNet-18 as the encoder and the use of free-tier computing. Both follow from the limited resources of the project, and the choice of encoder is examined further by the comparison with other model frameworks in Section 3.2.8.

3.2.2 Dataset and Label Construction

3.2.2.1 Source and Collection Protocol

All experiments use the public dataset of Valles-Coral et al.[1], released under a CC BY 4.0 licence. It was collected at the University Medical Center of the Universidad Nacional de San Martín, Peru, from 909 volunteers aged 18 to 25 (540 women and 369 men). Trained professionals (licensed nurses, nutritionists and general physicians) supervised the sessions. Hemoglobin was measured with the non-invasive Rad-67 device, and the visual

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recordings were made at the same time with a Samsung Galaxy A73 5G smartphone that was controlled remotely with the scrcpy software, so that lighting and capture angle were standardized across participants.

Three recordings were made for each participant, one for each visual site:

1. **Fingertip videos (yemas):** a 30-second recording of the right index fingertip at 60 frames per second, with fixed ISO 250, shutter speed 1/60 s, focus 0.0 and white balance 4400 K.

2. **Palm videos (palmas):** a recording in which the participant starts with a closed fist and progressively opens the hand.

3. **Nail photographs (uñas):** a photograph of the fingernails taken in a light box, with aperture f/1.8, shutter speed 1/180 s, ISO 50 and a 5.06 mm lens.

The fixed exposure and white balance matter because anemia screening from images relies largely on colour: constant camera settings make colour differences more likely to reflect the participants than the camera.

Each participant is also linked to clinical data: age, sex, hemoglobin level, heart rate, oxygen saturation, weight, height, and eight binary symptom indicators (fatigue, weakness, irregular heartbeat, shortness of breath, dizziness, chest pain, cold hands and feet, headache). The files are named with anonymous identifiers (ID001, ID002, ...) in place of the national identity number that was used internally by the data providers, and each class folder holds the three modalities in parallel directories. The published dataset contains only the original, unprocessed videos and photographs; frame extraction, cropping and resizing are left to the user[1]. They are therefore part of this thesis and are described in Section 3.2.3.

3.2.2.2 Classes and Labels

Participants are graded Normal, Mild (*Leve*) or Moderate (*Moderada*); no participant has severe anemia. The grades were not defined arbitrarily: the hemoglobin thresholds were set in consultation with the Blood Bank of the city of Tarapoto, which accounted for the local altitude and other regional physiological conditions[1]. They therefore differ from generic international cut-offs and should be understood as local clinical standards. Two label spaces are used in this thesis.

- **Binary:** Normal is the negative class (label 0) and Anemic, which merges Mild and Moderate, is the positive class (label 1). Binary labels were read from the processing table `Dataset2.csv`, which also contains the patient-level split assignment and the file paths of the three modalities. The binary label simplifies the task and corresponds to the screening decision (refer or do not refer). The cohort contains 623 Normal and 286 Anemic participants (211 Mild and 75 Moderate), roughly two Normal participants for each anemic one.

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- **Three-class:** Normal, Mild and Moderate, derived from the one-hot class columns of the participant metadata file `metadata.csv`. This label space preserves severity but contains a very small Moderate class.

Throughout the thesis, sensitivity refers to the recall of the Anemic class, so that a false negative is an anemic participant classified as normal.

3.2.2.3 Cohort Composition

Table 3.2 shows how the classes are distributed over the sexes. The cohort is strongly unbalanced in two ways. Normal participants are about twice as numerous as anemic ones, and the Moderate class contains only 75 participants. In addition, anemia is concentrated in women: 46.85% of the women are anemic against 9.21% of the men, and only one man has moderate anemia.

Table 3.2: Distribution of the 909 participants by sex and severity[1].

Sex	Normal	Mild	Moderate	Anemic (%)
Women	287	179	74	46.85
Men	335	33	1	9.21

This composition has two consequences. Accuracy alone is a weak measure, since predicting Normal for everyone already scores about two thirds, which is why class-imbalance handling, sensitivity and macro-averaged scores are used (Section 4.2). And sex and age are themselves informative about the label, so a model that sees demographic variables can score well without extracting anything from the images; this is why the fusion ablation includes a demographic-only reference (Section 3.2.7).

3.2.2.4 Availability of Files and Data Splitting

Not every participant has all three recordings, and not every recording could be used. The dataset provides 909 participants per modality. The recordings that were found and processed number 826 nail photographs, 833 palm videos and 838 fingertip videos. The existence and readability of each file were then verified, and the verified numbers used in the single-modality experiments are given in Table 3.3. Subjects were included in the multimodal experiments only if all required modalities were available, which left 808 participants for the three-class fusion experiments.

Table 3.3: Verified files per class in the single-modality experiments.

Modality	Normal	Mild	Moderate	Total
Nail photographs (uñas)	519	164	59	742
Palm videos (palmas)	554	201	74	829
Fingertip videos (yemas)	558	201	75	834

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Data were split at the patient level with stratification into 70% training, 15% validation and 15% test subsets, so that no participant appears in more than one subset and the class balance is similar in each subset. Patient-level splitting is essential for the video modalities, because frames of the same video are almost identical and would otherwise leak between training and test data. For the nail modality this gave 359 Normal and 158 Anemic training images, 81 Normal and 34 Anemic validation images, and 79 Normal and 31 Anemic test images. The multimodal three-class experiments use a fixed test set of 123 participants (87 Normal, 29 Mild and 7 Moderate) that is shared by all three-class experiments; the binary fusion experiment uses a test set of 110 participants (Section 3.2.6).

3.2.3 Preprocessing

The dataset provides only the original recordings, which were converted into the fixed-size images that the network takes as input. The modalities pose different problems: a fingertip video fills the frame but is long and redundant, a palm video contains a closed-to-open transition of which only part is informative, and in a nail photograph the nail occupies a small part of a whole-hand image. A separate procedure was therefore defined for each modality, following a small set of common rules.

3.2.3.1 Common Design Principles

- **One procedure for every class.** Normal, Mild and Moderate recordings pass through exactly the same code, with no class-specific branches. This rule was adopted after an earlier run in which one class was left at the raw capture resolution while the others were resized, which would have made image size correlate with the label.
- **Fixed output size.** Every saved image or frame is resized to 224×224 pixels with area interpolation (OpenCV `INTER_AREA`), which limits aliasing when downscaling, and stored as JPEG at the OpenCV default quality of 95. The resize is square and does not preserve the aspect ratio, so non-square source frames are slightly stretched.
- **Identifier preserved.** Output files keep the participant identifier (for example ID009), so every frame can be traced to its participant and splits can be made per participant.
- **No normalization stored.** The saved files are plain resized images. The ImageNet mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$ required by the pretrained backbone, and the augmentation described in Section 3.2.5, are applied during training.

3.2.3.2 Fingertip Videos

By the recording protocol the fingertip covers the whole camera lens, so the frame content is the fingertip itself and no localization or cropping is needed. The only decisions are

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which frames to keep and how many. The first and the last 10% of the frames of each clip are discarded, because the finger is being placed at the start and lifted at the end. Thirty-two frames are then taken at evenly spaced positions from the remaining 80%, which gives uniform coverage of the steady part of the recording, and each frame is resized to 224×224. The trimming is defined as a percentage of each clip's own length, so clips of different durations are shortened in proportion. A clip with fewer than 32 frames would have kept all of its frames, but no such case occurred.

3.2.3.3 Palm Videos

In the palm video the participant starts with a closed fist and opens the hand, so the early part of each clip shows a hand that is still moving into position. The first 25% and the last 10% of each clip are discarded, and 16 evenly spaced frames are taken from the remaining 65% and resized to 224×224. Fewer frames are kept than for the fingertip because consecutive palm frames differ little once the hand is open.

3.2.3.4 Nail Photographs

The nail photographs needed the most work, because the nail occupies only a small part of the image. The raw photograph (2992×2992 pixels) shows several fingertips on a dark surface, with no palm in view.

Two approaches that did not work. The first version of the pipeline cropped a fixed central fraction (75%) of the photograph. The hand is not guaranteed to be at the centre, so many crops contained mostly background with the nail in a corner (for example participant ID104). A pretrained hand-landmark detector was then tested and reported no hand in any of the sample images, even when a hand was clearly visible. This was not a software fault: the detector expects a full hand with the palm visible, whereas these photographs show only fingertips.

Skin-colour localization. Because the hand is the only skin-coloured object on a dark and fairly uniform surface, it can be found by colour alone, without any trained model. The image is converted to the YCrCb colour space, which separates luminance from chrominance and so keeps skin tones in a compact range under changing brightness, and pixels with $\mathrm{Cr} \in [133, 173]$ and $\mathrm{Cb} \in [77, 127]$ are marked as skin. The mask is cleaned by a morphological opening with a 5×5 kernel, which removes isolated specks, and a closing with a 15×15 kernel, which fills small holes inside the finger. The largest connected region is kept and is accepted only if it covers at least 1% of the image. Its topmost point is taken as the nail location, because in these photographs the target finger extends furthest up in the frame.

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Cropping. A 900 × 900 pixel square window is centred horizontally on the detected point and shifted upward by 108 pixels (12% of the window). The shift is needed because the detected point is where visible skin begins, while the pale nail tip often lies outside the skin colour range and above that point. Without the shift the tip was cut by the top edge of the crop, as seen in several Moderate-class images (for example ID005, ID010, ID014 and ID044). The window is clamped to the image borders and the crop is resized to 224×224. A fallback to the central crop was defined for images in which no skin region is found, with the image flagged for manual review. It was never triggered: a skin region was found in all 826 photographs.

The skin thresholds were checked by visual inspection of the original photographs next to their crops (five images per class) and not by a quantitative evaluation of detection accuracy. Colour thresholds of this kind can be sensitive to lighting and skin tone, so this is a limitation of the procedure (Section 4.3.7).

3.2.3.5 Integrity Checks and Output

The recordings are stored on Google Drive, where writing many small files can fail silently: in an early run the script reported 203 of 203 videos processed while Drive showed only 175. Each file was therefore copied to local disk, processed there, copied back and verified (by frame count, or by byte size for images) before the next was started. Corrupted inputs and copy mismatches were logged, and none were found. Interrupted sessions resumed from the last completed participant, and a final check confirmed the expected frame count and image size for all fingertip and palm folders and the presence of every nail image. Table 3.4 summarizes the settings and the resulting data.

Table 3.4: Preprocessing settings and output for each modality.

	Fingertip (yemas)	**Palm (palmas)**	**Nail (uñas)**
Input	video	video	photograph
Discarded (start / end)	10% / 10%	25% / 10%	none
Frames per participant	32	16	1
Localization	none (finger fills lens)	none	skin colour in YCrCb; 900 × 900 crop
Output size	224 × 224	224 × 224	224 × 224
Participants processed	838	833	826
Output files	26,816	13,328	826

3.2.3.6 Preprocessed Inputs

Figure 3.3 shows the preprocessed inputs of one participant, one row per modality.

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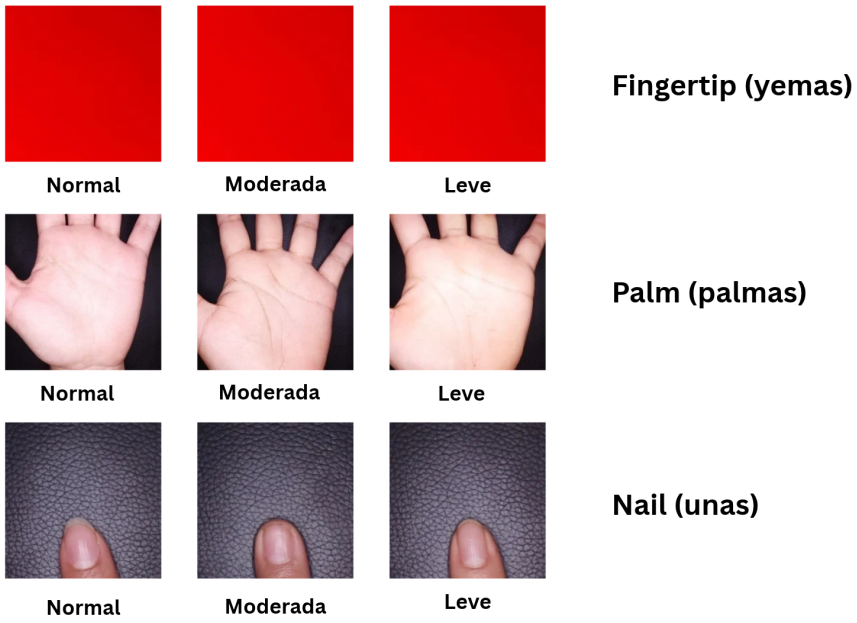


Figure 3.3: Preprocessed samples of the three modalities for one participant (all 224×224): eight of the 32 fingertip frames, eight of the 16 palm frames and the nail crop. Images are from the public dataset[1] (CC BY 4.0).

3.2.3.7 Limitations of the Preprocessing

The square resize stretches non-square frames, the nail localization depends on colour thresholds that were verified visually and not measured, the modalities have slightly different numbers of participants (so multimodal experiments use only participants with all required recordings, Section 3.2.2), and the trimming percentages and frame counts were fixed in advance and not tuned (their effect is examined in Section 3.2.5).

3.2.4 Single-Modality Baselines

The first step toward answering RQ1 is to determine what each modality can achieve on its own. For each modality, an ImageNet-pretrained ResNet-18 was fine-tuned, with its final fully connected layer replaced by a single output neuron that produces the logit of the Anemic class. Training used the binary cross-entropy with logits loss (`BCEWithLogitsLoss`) with class-imbalance weighting, the AdamW optimizer with a cosine annealing learning-rate schedule, a batch size of 32 and 10 epochs. After every epoch the model was evaluated on the validation set, and the checkpoint with the best validation score was kept and evaluated once on the test set. For the video modalities, the individual frames of a video are the training samples of the baseline, so its predictions are made frame by frame.

3.2.5 Ablation Study of Frame Sampling and Training Strategy

The baseline treats frames independently and uses a single sampling density and one training recipe. To answer RQ2, a controlled ablation varied two factors while the rest of

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the pipeline was fixed.

Factor 1: sampling density. Eight, 16 or 32 frames were sampled uniformly from every video, giving low, medium and high temporal density. A lower density reduces redundancy and inference cost, whereas a higher density may reduce noise at the expense of redundancy. The nail modality is a single photograph, so this factor does not apply to it.

Factor 2: training strategy. Three strategies were compared. Work 1 applies class-imbalance handling only (a weighted loss). Work 2 applies data augmentation only; the augmentation variants were the original frame and brightness, darkness and flip variants, applied per frame within a video, chosen to avoid changes that would alter the clinical color information. Work 3 combines both.

Model. Each frame is encoded by the pretrained ResNet-18, and the frame features are aggregated by mean pooling into one video-level vector, which passes through fully connected layers with batch normalization and dropout to produce the probability of anemia. A tuned decision threshold converts the probability into a binary prediction. Models were trained for 20 epochs with `BCEWithLogitsLoss`, and the checkpoint with the best validation AUC-ROC was kept.

3.2.6 Multimodal Fusion Framework

The single-modality experiments of the previous sections show what each recording can do alone. RQ3 and RQ4 ask the next question: whether the three recordings, and optionally the demographic and symptom variables, become more useful when they are combined, and how they should be combined. This section describes the framework that was built to answer it. The framework is shown in Figure 3.4 in the configuration that the ablation study later selected (Section 4.3.4); every other configuration studied in this thesis is obtained from it by exchanging one component, as explained in Section 3.2.7.

Following the figure from top to bottom, every visual modality first passes through its own ImageNet-pretrained ResNet-18, which is fine-tuned together with the rest of the model. The 512-dimensional average-pooled feature of the network is projected by a linear layer to a 256-dimensional embedding. The two video modalities are encoded frame by frame, with 32 frames for the fingertip and 16 frames for the palm, so each video yields a sequence of 256-dimensional vectors. The nail photograph is a single image and yields one vector directly, which is why the figure shows no temporal axis for it.

The sequences of the two videos must be reduced to one vector each before the modalities can be combined, and two ways of doing so are compared. Mean pooling simply averages the frame embeddings and ignores their order. The bidirectional LSTM (BiLSTM) instead reads the frames forwards and backwards with one layer of 128 hidden units

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per direction, concatenates the two final hidden states and projects them back to 256 dimensions, so that it can in principle exploit the order of the frames.

Some experiments add a fourth source of information. In these, twelve demographic and symptom features (age, heart rate, weight, height, sex and seven symptom flags: fatigue, weakness, irregular heartbeat, shortness of breath, dizziness, chest pain and headache) are scaled with statistics from the training data only and passed through a small network (linear $12 \rightarrow 256$, ReLU, dropout 0.3, linear $256 \rightarrow 256$). The result is a 256-dimensional embedding that is treated like the embedding of a fourth modality.

The embeddings are then fused, and the choice of the fusion module is the central design question of the thesis. In concatenation, the embeddings are placed side by side (768 numbers, or 1024 with the demographic embedding) and handed to the classifier, so that any interaction between the modalities has to be learned by the classifier itself. In self-attention fusion, each embedding becomes one token, a learned modality embedding is added to tell the tokens apart, and the tokens pass through one Transformer encoder layer (model dimension 256, four heads of 64 dimensions, feed-forward size 512, dropout 0.3, residual connections and layer normalization). Each modality can thereby weight the others before the output tokens are averaged into one 256-dimensional patient vector.

Finally, a small classification head turns the fused vector into a prediction. For concatenation it is a linear layer to 128 units, a ReLU, dropout of 0.3 and a linear layer to three logits; for self-attention it is a linear layer from 256 to 64 units, a ReLU, dropout of 0.3 and a linear layer to three logits, followed by a softmax at inference. The binary reference experiment uses one output logit instead of three.

3.2.7 Ablation Study of the Fusion Framework

A framework with several interchangeable parts raises an obvious question: which of the parts actually matter? An ablation study answers it by changing one component at a time, keeping the rest fixed, and measuring how the performance moves. Here the ablation has a second purpose, because it also has to decide which configuration is worth carrying forward to a wider comparison with other model frameworks.

All experiments share the same backbone and input pipeline, so that any difference in the results can be traced to the component that was changed. Every experiment uses 32 fingertip frames, 16 palm frames and one nail image, each resized to 224×224 pixels with ImageNet normalization and without data augmentation. Four components are varied: the fusion module (concatenation or self-attention), the temporal pooling (mean or BiLSTM), the demographic and symptom features (absent or present) and the task (three severity classes or binary Normal versus Anemic). The three-class experiments cover every combination of fusion, pooling and demographics, which gives eight configurations. Throughout the thesis, a difference between two results is given in percentage points (pp) and is always the modified model minus the baseline.

Table 3.5 lists the experiments. The three-class experiments use the 808 participants

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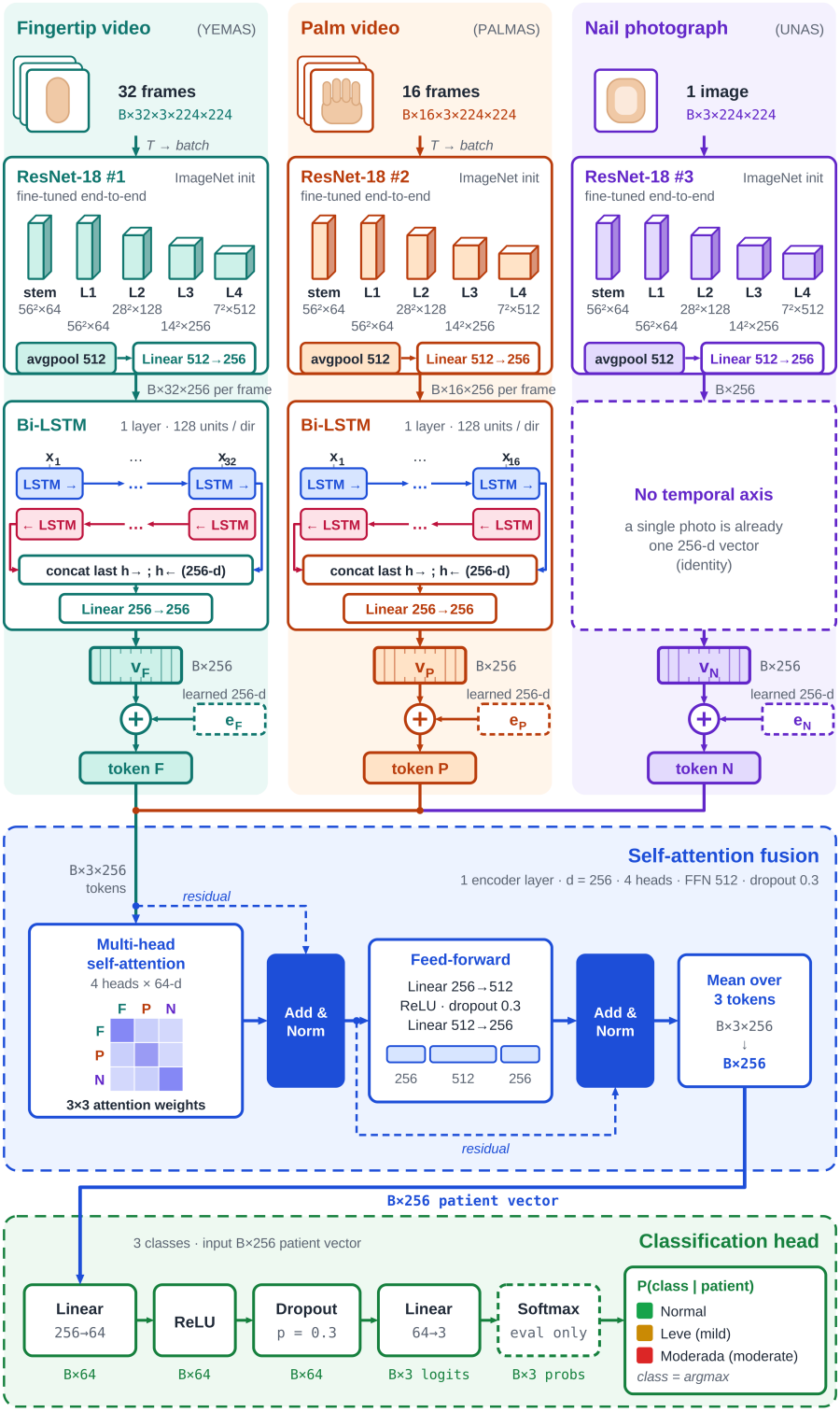


Figure 3.4: Architecture of the selected model: three ResNet-18 encoders, bidirectional LSTM pooling for the two video modalities, self-attention fusion of three modality tokens and a three-class classification head. Tensor shapes are given for a batch of size B. The other ablation configurations replace the BiLSTM by mean pooling, the self-attention block by concatenation, or add a fourth token for demographic and symptom variables.

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who have all three image modalities, with labels taken from the participant metadata file. The binary experiment, Experiment 0, stands apart: it uses a different cohort and a different label source, and it is therefore kept as a reference for the binary formulation rather than as the baseline of the three-class ablation (Section 4.3.3.6).

Table 3.5: Experiments of the fusion ablation. Every three-class experiment was run with both fusion modules. †The five-fold result of Experiment 3 with BiLSTM was found unreliable (Section 4.3.3.7).

Experiment	Task and test set	Pooling	Demographics	Protocols run
Ex-0	Binary, 110 participants	Mean	No	Single split; five-fold
Ex-1 (mean)	Three-class, 123 participants	Mean	No	Single split; five-fold
Ex-1 (BiLSTM)	Three-class, 123 participants	BiLSTM	No	Single split; five-fold
Ex-2 (supplementary)	Three-class, 123 participants	Mean	Yes	Five-fold only, AdamW
Ex-3 (mean)	Three-class, 123 participants	Mean	Yes	Single split; five-fold
Ex-3 (BiLSTM)	Three-class, 123 participants	BiLSTM	Yes	Single split; five-fold†

Every three-class configuration was evaluated under two protocols, because a single training run on a small dataset can be misleading. In the single-split protocol, a model was trained for 20 epochs on 562 participants, its checkpoint was chosen on 123 validation participants, and it was scored once on the 123 test participants. In the five-fold protocol, the 685 training and validation participants were pooled and divided once into five patient-level stratified folds of 548 training and 137 validation participants. Each fold trained for 10 epochs, its best checkpoint was chosen on its own validation part, and each of the five fold models was then scored on the same fixed test set. The five results are reported as mean ± standard deviation, and the five models are also combined by averaging their predicted probabilities, which gives the fold-averaged result. The test set is identical in both protocols, with 87 Normal, 29 Mild and 7 Moderate participants, and was never used for training or checkpoint selection. Because the protocols differ in training length and training-set size, results are compared within a protocol and never across protocols.

The remaining settings are common to all runs. Training used two Tesla T4 GPUs, a batch size of 16, dropout of 0.3 and a random seed of 42 that was set once at the start of each notebook. Experiments 1 and 3 used the Adam optimizer with a learning rate of 10^{-4}, and Experiment 2 used AdamW with the same rate and a weight decay of 10^{-4}. Because the Moderate class is rare, the three-class loss is a cross-entropy with class weights $w_c = N/(K\,n_c)$, where N is the number of training participants, $K = 3$ the number of classes and n_c the number of participants in class c. For the single split the weights are 0.4996, 1.3575 and 3.8231 for Normal, Mild and Moderate, and in the five-fold protocol they are recomputed within each fold. The binary experiment uses an unweighted binary cross-entropy. Checkpoints are selected by validation macro-F1 in the three-class runs and

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by validation ROC-AUC in the binary runs.

To judge how much the images add, two tabular baselines were trained on the twelve demographic and symptom features alone and scored on the same test set: a multinomial logistic regression and a random forest of 300 trees, both with balanced class weights. If an image-based model cannot clearly beat them, the images have not yet proven their value.

Two features of the design limit what the comparisons can show, and they should be kept in mind when the results are read. First, the fusion comparison changes more than the fusion rule: the classification head has 128 hidden units for concatenation and 64 for self-attention, and the fused vector has 768 or 1024 numbers for concatenation but only 256 for self-attention, so a difference between the two modules cannot be credited to the fusion rule alone. Second, Experiment 2 and the mean-pooling run of Experiment 3 use the same inputs but different optimizers, so they are not strict replicas of each other.

3.2.8 Comparative Benchmarking Protocol

The ablation studies compare variants of one framework, so they show which components of the selected model matter but not whether the framework is competitive with methods proposed elsewhere. The comparative benchmarking answers this complementary question. Eleven published methods, labelled B03 to B13, were re-implemented on the same three-class task (Normal, Mild and Moderate) and compared with the selected model of Section 4.3.3, which uses images only (fingertip video, palm video and nail photograph), BiLSTM pooling and self-attention fusion. The methods were re-implemented and not quoted, because published scores come from other data, mostly from binary tasks, and in some papers from splits made after augmentation (accuracies of 0.65 to 0.99), so they cannot be placed next to a three-class result. The protocol was fixed before any score was seen.

3.2.8.1 Benchmarked Methods

Each paper was chosen because it challenges one design decision of the selected model, so even a weaker paper says something about that decision. The papers form four families.

The first family concerns the *fusion strategy*. B04 re-implements the intermediate fusion of a multi-modal pneumonia classifier (ICECTE 2026)[67] with a MobileNetV3-S backbone. B05 applies Attentional Feature Fusion (Dai et al., 2021)[68] on ResNet-18 and is the closest rival to the self-attention fusion of the selected model. B06 follows the fusion of CT images and health records of Huang et al. (2020)[69] by late averaging of an image model and a tabular ElasticNet. B10 is a hybrid CNN-Transformer with mutual distillation (MV-HFMD)[70], run with one frame from each of the same three views. B12 takes the RCBAM-MobileNetV2 image branch of a paper that fuses health records with conjunctiva images (Ramzan et al., 2024)[15], tested on the nail photograph alone.

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The second family concerns *nail and colour features*. B03 re-implements a multi-input network for anemia detection from nail images (ICBSII 2024)[4], an EfficientNet-B4 with an a^* colour-statistics branch. B07 follows Asare et al. (2023)[8] with a k-nearest-neighbour classifier ($k = 5$) on 21 CIELAB colour features as a classical baseline. B11 follows Navarro-Cabrera et al. (2025)[13] and asks whether one nail photograph with a pretrained DenseNet169 is enough.

The third family concerns *hemoglobin estimation from body-part images*: B09 adapts BPANet (Lin et al., 2025)[6] with a ResNet-50, age, sex and a hemoglobin head, and B13 applies the EfficientNet-B0 with GeM pooling of Kato et al. (2024)[12] to the nail photograph. The fourth family is *ensembling*: B08 follows Murugeswari et al. (2025)[10] with a majority vote of three ResNet-18 branches, to be compared with the single fused model of this thesis.

The runs differ in how directly they can be compared with the selected model. B03 to B06 are direct comparisons, with the same inputs and the same protocol. B07, B08, B09 and B13 are partial, because their input or scoring differs. B10, B11 and B12 are not direct, because they address a different task or input, and all three are single runs.

3.2.8.2 Who Ran Each Method and Protocol Groups

Every method was re-run by a member of the team, and each run is stored with its notebook, metrics, confusion matrix and source paper, so no score is taken from the literature. The first member ran B03 to B08, which share one code base and the protocol of the selected model. The second member ran B09 to B11, and the third member ran B12 and B13. The members are called first, second and third in the order of their runs.

The runs fall into three groups, which are marked in every table and figure of the results. B03 to B08 follow the protocol of the selected model and are scored on the same 123 test participants (87 Normal, 29 Mild and 7 Moderate). B13 is scored by five-fold cross-validation over all 823 nail participants, so its folds contain the test participants of the selected model. B09 to B12 are single runs on their own test sets (87/29/7, 82/30/11, 83/30/11 and 165/61/22 participants per class); B09 to B11 use the splits of the second member and B12 a stratified random 70/30 split of the nail photographs. A gap between a single run and the selected model therefore mixes the effect of the method with that of the split and the random seed.

3.2.8.3 Design Decisions of the Benchmark

Task and metrics. The three-class task of the selected model was kept, because merging classes would hide the Moderate class, where the methods differ most. Macro-F1 is the primary metric and macro-AUC the second, with accuracy only as context: always predicting Normal gives an accuracy of 0.707, so accuracy rewards ignoring anemia, whereas macro-F1 weights the three classes equally and the AUC does not depend on a threshold.

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One run per paper, chosen in advance. Only the variant that each paper proposes or reports as its headline method was run, so that the benchmark does not take the best of several variants per paper, which would inflate its scores relative to the single selected model.

The protocol of the selected model where possible. B03 to B08 use the same inputs, the same five patient-level folds over 685 participants and the same fixed test set as the selected model, with 10 epochs, Adam at 10^{-4} and the checkpoint chosen by validation macro-F1. A gap therefore reflects the method and not the split, and results are the mean and standard deviation over the five fold models. The other runs could not be placed on this split; they are kept and marked, but they have no standard deviation or significance test, and only large gaps are interpreted. The paired permutation test over folds was applied to B03 to B08 only, because with five folds the smallest attainable p is 0.0625.

Nail-only runs. For B12 and B13 only the nail photograph is reported. The nail set holds almost one image per participant, so an image-level split is effectively participant-level and cannot leak between training and test data. For B12, only the image branch is reported, because the notebook audit found three one-hot columns of the class label in the health-record table, so the 97 to 99.9% accuracy of the record-only and fusion experiments reflects target leakage.

Overlapping reproductions of B07. The run under the protocol of the selected model was kept, because only it scores the same 123 participants and has a fold-to-fold spread. A second, independent reproduction on a 70/15/15 split reached a macro-F1 of 0.354 for k-NN on the nail images and does not change the conclusion.

The benchmark did not measure parameter counts or training time, so computational cost is not compared, and clinical suitability is not assessed.

3.3 Project Plan

This section records the schedule of the project, and Section 3.4 the division of work. The initial plan (Tables 3.6 and 3.7) was revised as the work progressed: explainability and an XGBoost-based integration were listed in the original schedule but were not pursued, and the multimodal stage focused on ResNet-18 fusion with an encoder for demographic and symptom variables.

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Table 3.6: Project plan and task allocation (weeks 1–12).

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Define problem	20%	40%	60%	80%	100%	-	-	-	-	-	-	-
Literature review	10%	30%	50%	70%	90%	100%	-	-	-	-	-	-
Data collection / verification	10%	25%	40%	55%	70%	85%	100%	-	-	-	-	-
Baseline model (nail / fingertip / palm)	-	-	10%	25%	40%	60%	80%	100%	-	-	-	-
Model training	-	-	-	10%	25%	40%	60%	80%	100%	-	-	-
Validation	-	-	-	-	10%	25%	40%	60%	80%	100%	-	-
Testing	-	-	-	-	-	10%	25%	40%	60%	80%	100%	-
Fine-tuning	-	-	-	-	-	-	10%	25%	40%	60%	80%	100%
Evaluation metrics	-	-	-	-	-	-	-	10%	30%	50%	75%	100%
Report writing	-	-	-	-	-	-	-	-	10%	30%	60%	100%
Project submission	-	-	-	-	-	-	-	-	-	-	50%	100%

Table 3.7: Project plan and task allocation (weeks 13–24).

Task	W13	W14	W15	W16	W17	W18	W19	W20	W21	W22	W23	W24
Multimodal model development	10%	25%	40%	60%	80%	100%	-	-	-	-	-	-
XGBoost integration	-	10%	25%	40%	60%	80%	100%	-	-	-	-	-
MLP metadata integration	-	-	10%	25%	40%	60%	80%	100%	-	-	-	-
Explainable AI implementation	-	-	-	10%	25%	40%	60%	80%	100%	-	-	-
Model optimization	-	-	-	-	10%	25%	40%	60%	80%	100%	-	-
Final performance improvement	-	-	-	-	-	10%	30%	50%	70%	90%	100%	-
Extended analysis and documentation	-	-	-	-	-	-	10%	30%	50%	70%	90%	100%

3.4 Task Allocation

Literature review (all members). Each member read five research papers and summarized their methods, findings and relevant techniques.

Engineering problem solving (all members). The members collaborated on model design, preprocessing and integration problems.

Dataset-specific tasks.

- Tasmim Jahan Rimvy: nail (uñas) dataset preprocessing and validation.
- Jasmin Sultana Shimu: fingertip (yemas) dataset preprocessing and validation.
- Md Fazley Rabbi: palm (palmas) dataset preprocessing and validation.
- Sayma Sultana: testing and evaluation across all datasets, final evaluation metrics and confusion matrices.

Experimental tasks.

- Tasmim Jahan Rimvy: nail dataset experiments.
- Jasmin Sultana Shimu: fingertip dataset experiments.
- Md Fazley Rabbi and Sayma Sultana: palm dataset experiments.

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3.5 Summary

This chapter set out the design of the study. The requirements and the system context frame the model component that is evaluated (Section 3.1). The methodology (Section 3.2) justifies the main design decisions against their alternatives and defines the dataset and labels, the preprocessing, the single-modality baselines, the tri-modal fusion framework, the two ablation studies and the protocol of the comparative benchmarking. Single-modality ResNet-18 baselines address RQ1, an ablation of sampling density and training strategy addresses RQ2, and a component-wise ablation of fusion module, temporal pooling and demographic features addresses RQ3 and RQ4 and selects the model that is analysed in detail. The benchmarking protocol compares that model with eleven re-implemented published methods and addresses RQ5. The project plan and the task allocation record how the work was organized. Chapter 4 defines the evaluation protocol that governs all experiments and reports the results.

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Chapter 4

Implementation and Results

Chapter 3 defined the experiments that address the research questions. This chapter reports how they were implemented and what they showed. It first describes the environment in which the experiments were run (Section 4.1) and the protocol by which every model was tested and evaluated (Section 4.2). The results are then presented and discussed in the order of the research questions (Section 4.3), from the single-modality baselines (RQ1) through the ablation of frame sampling and training strategy (RQ2) to the fusion experiments (RQ3 and RQ4), the selected model and the comparative benchmarking (RQ5), and are interpreted together in Section 4.3.6. Following the reporting policy of the thesis, the single-modality ablation is summarized by the best configuration for each modality.

4.1 Environment Setup

The single-modality experiments were run on Google Colab and a subset on Kaggle Notebooks; the fusion experiments were run on Kaggle Notebooks with two NVIDIA Tesla T4 GPUs (data-parallel training with a batch size of 16, or 8 on a single GPU). The software environment is listed in Tables 4.1 and 4.2. All models were implemented in PyTorch.

Component	Details
Operating system	Windows 11 and Ubuntu 22.04 LTS
Python	3.12.13
GPU	NVIDIA Tesla T4
GPU memory	15.64 GB
CUDA	12.8
PyTorch	2.10.0+cu128

Table 4.1: System environment (Google Colab and Kaggle, Tesla T4 GPU).

The free-tier resources limited session length and GPU memory, which restricted the number of hyperparameter settings that could be explored; this is discussed in Section 4.3.7.

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Library	Version
NumPy	2.4.6
Pandas	2.3.3
Scikit-learn	1.6.1
Matplotlib	3.10.0
Seaborn	0.13.2
SciPy	1.16.3

Table 4.2: Python libraries and versions.

4.2 Testing and Evaluation

All models are tested on held-out data and scored with the same metrics, so that results are comparable wherever the design allows. This section defines the metrics, the rules for model selection and test data, the unit of evaluation, the conditions under which results may be compared, the statistical test and the uncertainty of the results.

Metrics. In the binary setting the Anemic class is positive. Accuracy, sensitivity (recall of the Anemic class), specificity, precision, F1-score and the area under the ROC curve (AUC-ROC) are reported. In the three-class setting, accuracy, macro-averaged precision, recall and F1-score, the macro-averaged one-versus-rest AUC-ROC, the per-class precision, recall and F1 and the confusion matrix are reported. Macro-F1,

$$\text{macro-F1} = \frac{1}{3}\sum_{c=1}^{3} F1_c, \qquad F1_c = \frac{2P_cR_c}{P_c + R_c}, \tag{4.1}$$

is the main metric and also the checkpoint-selection criterion. It gives the small Moderate class the same weight as the large Normal class and therefore exposes failures that accuracy hides. Accuracy needs particular caution: predicting Normal for every test participant already gives $87/123 = 0.707$.

Model selection and test data. Checkpoints are selected on the validation set. Test results are reported for the selected checkpoint.

Unit of evaluation. Fusion experiments evaluate one prediction per participant. The single-modality baseline for the video modalities predicts per frame, and its confusion matrices therefore count frames (16 per video).

Comparability. Because the label space, the test set and the protocol differ between groups of experiments (Table 3.5), accuracy values are comparable only within a group. In particular, binary and three-class results are not directly comparable, and neither are single-modality and fusion results.

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Statistical test. Fusion modules are compared with a paired sign-flip permutation test on the five per-fold macro-F1 values (2000 random sign patterns). With five pairs there are only $2^5 = 32$ sign patterns, so the smallest attainable two-sided p-value is $2/32 = 0.0625$ and the test cannot reach $p < 0.05$ for any outcome. Its outcome is therefore used only as a rough guide.

Uncertainty. With test sets of 110 to 126 participants, a 95% confidence interval for an accuracy of about 70% extends roughly ±8 to 9 percentage points, and the Moderate class has only seven test participants. Differences between configurations that are smaller than this should be regarded as suggestive. The standard deviation over folds reflects changes in the training and validation data but not the sampling uncertainty of the small test set. Where a single value is reported for the best of several configurations, it is also subject to selection effects.

4.3 Results and Discussion

This section reports the results for each research question in turn and then discusses them together. Each subsection states the question it answers, the evidence, and what the evidence does and does not support.

4.3.1 Single-Modality Baselines (RQ1)

The baselines establish how much information each modality carries when only a pretrained ResNet-18 and a simple training recipe are used. Table 4.3 shows the test results. The fingertip and palm models reach almost identical accuracy of about 66%, with specificity above 71% and sensitivity of about 55%, and the nail model has the lowest accuracy and the lowest AUC-ROC. In all three cases the area under the ROC curve (0.62 to 0.67) shows that the models rank anemic participants above normal ones only moderately better than chance.

Modality	Accuracy	AUC-ROC	Sensitivity	Specificity	F1
Nail (uñas)[†]	56.36	62.03	51.90	67.74	63.08
Palm (palmas)	66.35	67.30	54.73	71.24	49.05
Fingertip (yemas)	66.37	65.57	55.64	71.54	51.85

Table 4.3: Test performance (%) of the single-modality baselines. Palm and fingertip results are computed per frame.

[†] The nail values of sensitivity, specificity and F1 as reported refer to the Normal class as positive class: they are reproduced exactly by 41 of 79 Normal and 21 of 31 Anemic test images being classified correctly. With the Anemic class as positive, as in the other rows, these values are sensitivity 67.74%, specificity 51.90% and F1 46.67%.

The confusion counts of the two video baselines show where their errors lie. For the palm model, 1003 of the 1408 Normal frames were classified correctly and 405 were wrongly flagged as anemic, while 324 of the 592 Anemic frames were found and 268 were missed.

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For the fingertip model, 973 of the 1360 Normal frames were classified correctly and 387 were wrongly flagged, while 365 of the 656 Anemic frames were found and 291 were missed. In both models, therefore, most Normal frames are recognized (about 71%) but 44 to 45% of the Anemic frames are missed, so the dominant error is the false negative, which is the more costly error in a screening setting. About 29% of the Normal frames are flagged as anemic, which would lead to unnecessary referrals. The two models behave almost identically, in line with their near-identical accuracy in Table 4.3.

Because the baseline treats each frame as an independent sample and uses one sampling density and one training recipe, these results are best read as a reference point. They answer RQ1 in the following way: each modality contains a detectable but weak signal under this recipe, with no modality clearly superior in accuracy.

4.3.2 Effect of Frame Sampling and Training Strategy (RQ2)

The ablation examined 8, 16 and 32 sampled frames with Work 1 (class-imbalance handling), Work 2 (augmentation) and Work 3 (both) for the video modalities, and Work 1 to 3 for the nail photographs. Table 4.4 compares the best configuration of each modality with its baseline.

Modality	Configuration	Accuracy	AUC-ROC	Sensitivity
Nail	Baseline	56.36	62.03	67.74‡
	Best: Work 3 (imbalance + augmentation)	**67.27**	**71.23**	61.29
Palm	Baseline (frame level)	66.35	67.30	54.73
	Best: 8 frames, Work 1 (imbalance)	**67.20**	**68.90**	67.00
Fingertip	Baseline (frame level)	66.37	65.57	55.64
	Best: 32 frames, Work 2 (augmentation)	**73.87**	**74.27**	61.29

Table 4.4: Best configuration of each modality compared with its baseline (test set, %). Sensitivity is the recall of the Anemic class. ‡Equivalent of the reported Normal-class recall of 51.90% (see Table 4.3).

The changes in accuracy and AUC-ROC relative to the baseline are +10.9 and +9.2 percentage points for the nail, +0.9 and +1.6 points for the palm, and +7.5 and +8.7 points for the fingertip. Three observations follow.

First, the fingertip video benefits most from a dense sampling (32 frames) combined with augmentation and reaches the highest unimodal accuracy of 73.87% and AUC-ROC of 0.743. This is consistent with the expectation that the temporal content of the fingertip recording carries useful information and that averaging more frames reduces noise.

Second, the palm model improves very little. Its best configuration uses the smallest number of frames and only imbalance handling, and the gain of 0.9 percentage points in accuracy is far smaller than the uncertainty of the test set. For the palm, the data therefore do not show a clear benefit of denser sampling or augmentation.

Third, the nail model shows the largest gain in accuracy and AUC-ROC but a lower sensitivity than the baseline once all rows are expressed with the Anemic class as positive. The improvement of the nail model therefore consists mainly of a better specificity and

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ranking performance and not of better detection of anemic participants. Because the best nail configuration combines weighted loss and augmentation, the gain cannot be attributed to either alone.

In answer to RQ2, sampling density and training strategy affected the modalities differently: the fingertip model gained substantially, the nail model gained in accuracy and AUC-ROC, and the palm model gained marginally. Because only the best configuration of each modality is reported, and the test sets are small, the size of these gains should be treated as an upper estimate (Section 4.3.7).

4.3.3 Fusion Ablation Results (RQ3 and RQ4)

The single-modality results show that each recording carries some signal. The question of this section is whether combining the recordings helps, and which design choices of the fusion framework make a difference. The results follow the design of Section 3.2.7. They are first read as a whole under the five-fold protocol, then compared with the single-split protocol, then taken apart component by component, and finally checked for reliability, because one result stands out in a way that needs explaining. All values are test-set results on the fixed test set of 123 participants unless stated otherwise, and differences are given in percentage points (pp) as modified model minus baseline. In the tables, Concat. stands for concatenation fusion and Self-att. for self-attention fusion.

4.3.3.1 The Overall Picture Under Five-Fold Evaluation

Table 4.5 shows the five-fold results, which are the most stable evidence available. The first impression is that the configurations are hard to tell apart. Six of them (the image-only and the mean-pooling demographic configurations, each with both fusion modules) can be compared fairly with one another, and their mean macro-F1 lies between 0.380 and 0.459 and their mean macro-AUC between 0.675 and 0.721. The fold-to-fold standard deviation of macro-F1 is 0.040 to 0.089, so most gaps between these configurations are smaller than the spread inside a single configuration. Accuracy needs particular care here: the mean accuracy of 0.623 to 0.670 stays below the 0.707 that is reached by predicting Normal for every participant.

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Table 4.5: Three-class results under the five-fold protocol on the fixed test set (123 participants), mean ± standard deviation over five fold models. The tabular baselines are single fits scored once. †Not reliable (Section 4.3.3.7).

Configuration	Fusion	Macro-F1	Macro-AUC	Accuracy
Image-only, mean pooling (Ex-1)	Concat.	0.440 ± 0.089	0.711 ± 0.056	0.667 ± 0.048
	Self-att.	0.433 ± 0.040	0.708 ± 0.027	0.623 ± 0.058
Image-only, BiLSTM pooling (Ex-1)	Concat.	0.380 ± 0.041	0.675 ± 0.043	0.626 ± 0.063
	Self-att.	$\mathbf{0.459 \pm 0.075}$	$\mathbf{0.715 \pm 0.033}$	$\mathbf{0.670 \pm 0.040}$
Image + demographics, mean pooling (Ex-3)	Concat.	0.404 ± 0.053	0.721 ± 0.036	0.655 ± 0.054
	Self-att.	0.410 ± 0.065	0.692 ± 0.047	0.634 ± 0.038
Image + demographics, BiLSTM pooling (Ex-3)†	Concat.	0.668 ± 0.032	0.891 ± 0.027	0.878 ± 0.032
	Self-att.	0.687 ± 0.034	0.912 ± 0.032	0.836 ± 0.064
Image + demographics, mean pooling (Ex-2, supplementary)	Concat.	0.427 ± 0.077	0.711 ± 0.047	0.646 ± 0.052
	Self-att.	0.417 ± 0.057	0.718 ± 0.045	0.589 ± 0.050
Tabular logistic regression	—	0.407	0.717	0.480
Tabular random forest	—	0.344	0.699	0.699

The tabular baselines put these numbers in perspective. A logistic regression on the twelve demographic and symptom features alone reaches a macro-F1 of 0.407 and a macro-AUC of 0.717, and no comparable image-based configuration exceeds its macro-F1 by more than about five percentage points. The images therefore add, at most, a small amount to what the tabular features already offer under this setup.

One configuration breaks the pattern. The BiLSTM model with demographic features (marked with a dagger) reaches a macro-F1 of 0.668 and 0.687, about 26 to 28 pp above every other configuration. A jump of this size would be the most important finding of the thesis if it were real, which is why it is examined carefully in Section 4.3.3.7 before anything is concluded from it. Figure 4.1 previews the reason for doubt: this configuration classifies the Mild class almost perfectly, which no other configuration, not even the tabular logistic regression, comes close to doing.

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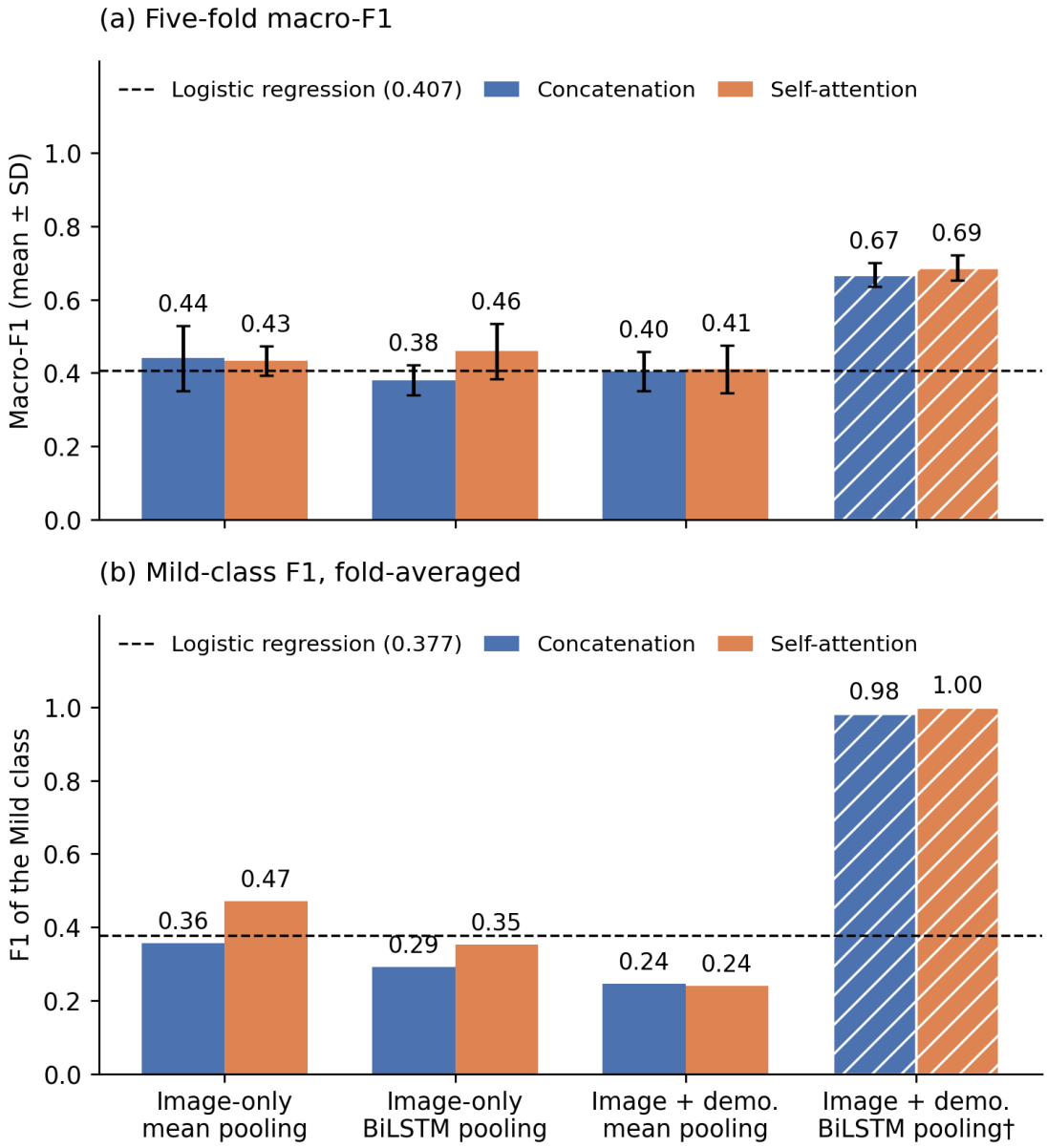


Figure 4.1: Five-fold test results of the eight three-class configurations, grouped by pooling and input. Hatched bars (†) are the unreliable configuration. (a) Macro-F1, mean over folds with one standard deviation. (b) F1 of the Mild class from the fold-averaged predictions. The dashed lines mark the tabular logistic regression.

4.3.3.2 Why Single Runs Cannot Be Trusted

If the five-fold results are blurred, the single-split results (Table 4.6) are contradictory, and the contrast is instructive. With image-only inputs and mean pooling, self-attention beats concatenation by 13.5 pp of macro-F1 in the single split (0.468 against 0.333), whereas under five-fold the two differ by only 0.7 pp, in favour of concatenation. With image-only inputs and BiLSTM pooling the order reverses: concatenation leads in the single split (0.439 against 0.406), while self-attention leads by 8.0 pp under five-fold. Adding demo-

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graphic features to the mean-pooling concatenation model changes macro-F1 by +10.8 pp in the single split but by −3.6 pp under five-fold. Differences of five to thirteen percentage points between single runs can therefore not be read as component effects. One training run and a test set with only seven Moderate participants are simply too noisy to support them, and for this reason the five-fold results, not the single splits, carry the conclusions below.

Table 4.6: Three-class results under the single-split protocol (20 epochs, one run per configuration, 123 test participants). †See Section 4.3.3.7.

Configuration	Fusion	Macro-F1	Macro-AUC	Accuracy
Image-only, mean pooling (Ex-1)	Concat.	0.333	0.659	0.455
	Self-att.	0.468	0.750	0.593
Image-only, BiLSTM pooling (Ex-1)	Concat.	0.439	0.742	0.569
	Self-att.	0.406	0.697	0.569
Image + demographics, mean pooling (Ex-3)	Concat.	0.440	0.677	0.626
	Self-att.	0.373	0.656	0.577
Image + demographics, BiLSTM pooling (Ex-3)†	Concat.	0.433	0.695	0.569
	Self-att.	0.512	0.746	0.691
Tabular logistic regression	—	0.417	0.711	0.496
Tabular random forest	—	0.323	0.698	0.650

4.3.3.3 Does Self-Attention Fusion Help?

Fusion is the one component that every experiment compares, so it offers the most evidence (Table 4.7). Among the three comparable three-class settings, self-attention has the higher mean macro-F1 in two. Only one of these gains is sizeable: with image-only inputs and BiLSTM pooling, self-attention improves macro-F1 by 8.0 pp and wins in four of the five folds. In the other two settings the difference is below one point, and in the supplementary Experiment 2 concatenation is ahead by 1.1 pp. In the binary task, self-attention raises the mean F1 by 12.0 pp, but the mean ROC-AUC does not change.

Table 4.7: Self-attention (SA) compared with concatenation under five-fold. Difference in the mean fold score in pp; p is the two-sided sign-flip permutation test on the five per-fold scores. †Not reliable.

Configuration	Metric	Diff. (pp)	Folds favouring SA	p
Image-only, mean pooling (Ex-1)	Macro-F1	−0.7	3/5	0.948
Image-only, BiLSTM pooling (Ex-1)	Macro-F1	+8.0	4/5	0.178
Image + demographics, mean pooling (Ex-3)	Macro-F1	+0.6	3/5	0.702
Image + demographics, mean pooling (Ex-2)	Macro-F1	−1.1	2/5	0.875
Image + demographics, BiLSTM pooling (Ex-3)†	Macro-F1	+2.0	3/5	0.548
Binary, mean pooling (Ex-0)	F1	+12.0	4/5	0.251
	ROC-AUC	−0.02	2/5	1.000

None of these differences is statistically supported, since the smallest p-value is 0.178. The test also has a built-in limit: with five folds there are only 32 possible sign patterns, so the smallest attainable two-sided p-value is $2/32 = 0.0625$, and a five-fold comparison of this kind cannot reach the 0.05 level however large the difference. The evidence therefore

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does not show that self-attention is better or worse than concatenation. It only suggests that self-attention may help when frames are pooled by a BiLSTM, a pattern that a larger evaluation would have to confirm. Because the two modules also differ in head size and fused-vector size, even a clear difference could not be credited to the fusion rule alone.

The per-class results add a detail (Table 4.10). For the image-only BiLSTM model, self-attention raised the Mild F1 from 0.291 to 0.353 and was the only image-only model that recovered any Moderate participant (F1 0.222, which is one correct participant out of seven). For image-only mean pooling, self-attention raised the Mild F1 from 0.357 to 0.471 but still found no Moderate participant.

4.3.3.4 Does the BiLSTM Help?

The second component is the way in which frames are pooled over time. For the image-only models, replacing mean pooling by the BiLSTM had opposite effects for the two fusion modules (Table 4.8): macro-F1 fell by 6.1 pp with concatenation and rose by 2.6 pp with self-attention. Both changes are smaller than the fold-to-fold standard deviation of the models involved (0.089 and 0.075), so the BiLSTM shows no consistent benefit over simple averaging.

A possible explanation is that the BiLSTM adds many parameters while only a few hundred participants are available for training. The training histories are consistent with this idea but do not prove it. In every reliable configuration the mean validation loss was lowest after the first or second epoch (about 0.98) and had risen to between 1.9 and 2.5 by epoch 10, while the training loss fell to between 0.15 and 0.24. The same overfitting appears with mean pooling, so it is not specific to the BiLSTM, and it also means that the epoch of the best checkpoint varies widely between folds.

Table 4.8: Change in the five-fold mean macro-F1 (pp) when one component is changed and the others are fixed. †The image data differ between the two compared runs, so the comparison is not fair.

Change made	Concat.	Self-att.	Fair comparison
BiLSTM instead of mean pooling, image-only	−6.1	+2.6	Yes
Add demographic data, mean pooling	−3.6	−2.3	Yes
BiLSTM instead of mean pooling, with demographics	+26.4	+27.7	No†
Add demographic data, BiLSTM pooling	+28.8	+22.8	No†

4.3.3.5 Do Demographic and Symptom Features Help?

The third component is the addition of the twelve demographic and symptom features. Under mean pooling, where the comparison is fair, they did not help: macro-F1 changed by −3.6 pp for concatenation and by −2.3 pp for self-attention, and macro-AUC moved by +1.0 and −1.6 pp, which is also within the fold spread. The supplementary Experiment 2, with the same inputs and a different optimizer, gave macro-F1 values of 0.427 and 0.417, close to the Experiment 3 values of 0.404 and 0.410, which shows that the result does not hinge on the optimizer.

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The tabular baselines explain why. Used alone, the features give a macro-F1 of 0.407 (logistic regression) and 0.344 (random forest). The random forest reaches an accuracy of 0.699, which is essentially the majority-class rate, because it finds none of the Moderate participants and few of the Mild ones. The features are therefore weak predictors on their own, and the experiments give no sign that the deep model gains from them. This does not show that the features are useless in general, only that they added nothing measurable under this setup.

4.3.3.6 Binary Versus Three-Class Formulation

The last component is the task itself. Table 4.9 reports the binary experiment. Under five-fold both fusion modules reach a ROC-AUC of about 0.724, which is similar to the three-class macro-AUC values, but the other metrics reveal a strong bias toward the majority class. With concatenation the mean sensitivity is only 0.200 and the specificity 0.927, so about one anemic participant in five is found, and the mean accuracy of 0.722 matches the share of Normal participants in the test set ($79/110 = 0.718$). Self-attention is less conservative, with a mean sensitivity of 0.406 and an F1 of 0.383, but it varies strongly from fold to fold. The unweighted loss is a likely contributor to this behaviour, although this was not tested.

Table 4.9: Binary Normal versus Anemic results (Ex-0, mean pooling, no demographics, unweighted loss). Sensitivity is the recall of the Anemic class. The test set has 110 participants (79 Normal, 31 Anemic). Five-fold values are mean ± standard deviation.

Protocol	Fusion	Acc.	Sens.	Spec.	F1	ROC-AUC
Single split	Concat.	0.727	0.484	0.823	0.500	0.737
	Self-att.	0.709	0.387	0.835	0.429	0.731
Five-fold	Concat.	0.722±0.038	0.200±0.183	0.927±0.027	0.263±0.185	0.724±0.067
	Self-att.	0.684±0.044	0.406±0.248	0.792±0.153	0.383±0.144	0.724±0.052

Experiment 0 cannot serve as the baseline against which the three-class models are ablated. It uses a different cohort (737 usable participants against 808) and a different test set (110 against 123 participants), and it reads its labels from `Dataset2.csv`, whose class column is misaligned for participant IDs 058 to 227 (Section 4.3.7). It is therefore reported as a reference for the binary formulation only, and no claim is made about whether the binary or the three-class formulation is easier.

4.3.3.7 Why the Strongest Result Is Not Trusted

Returning to the configuration that stood out in Table 4.5: the BiLSTM model with demographic features reaches a five-fold macro-F1 of 0.668 (concatenation) and 0.687 (self-attention) and a macro-AUC of 0.891 and 0.912, which is 26 to 28 pp above the same model with mean pooling. Three observations argue against taking this at face value.

The first concerns the data. This is the only five-fold notebook that reads its images from different input folders: the other notebooks read the frames and the nail images

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from one common input dataset, whereas this one reads fingertip, palm and nail images from three separately named datasets. A line-by-line comparison with the mean-pooling notebook of Experiment 3 shows that the pooling mode, the image folders and the name of the output folder are the only differences. The second concerns the error pattern, which is unlike that of every other run. The fold-averaged predictions classify all 29 Mild participants correctly with both fusion modules (Mild F1 of 0.983 and 1.000; Table 4.10), while no other three-class configuration exceeds 0.471 on this class and the demographic-only logistic regression reaches 0.377. The third is the single-split run of the same configuration: it used the common image folders and gave only 0.433 and 0.512 macro-F1 (Table 4.6), in line with all other single-split runs.

The available files do not reveal the cause. The preprocessing records mention an earlier version of the frame extraction in which one class was saved at raw capture resolution instead of the target size, so a class-dependent difference in the stored images is one possible explanation, but this is only a hypothesis that cannot be checked here. Until the experiment is repeated on the common image folders, this result is reported but excluded from every conclusion about pooling or demographic features, and it is marked with a dagger wherever it appears.

Table 4.10: Per-class F1 on the test set (support: 87 Normal, 29 Mild, 7 Moderate) from the fold-averaged predictions of the five-fold models, and recall of the Moderate class. †Not reliable.

Configuration	Fusion	Normal	Mild	Moderate	Mod. recall
Image-only, mean pooling (Ex-1)	Concat.	0.822	0.357	0.000	0.000
	Self-att.	0.793	0.471	0.000	0.000
Image-only, BiLSTM pooling (Ex-1)	Concat.	0.778	0.291	0.000	0.000
	Self-att.	0.828	0.353	0.222	0.143
Image + demographics, mean pooling (Ex-3)	Concat.	0.815	0.245	0.000	0.000
	Self-att.	0.783	0.240	0.167	0.143
Image + demographics, BiLSTM pooling (Ex-3)†	Concat.	0.925	0.983	0.143	0.143
	Self-att.	0.929	1.000	0.400	0.571
Tabular logistic regression	—	0.631	0.377	0.213	0.714
Tabular random forest	—	0.816	0.216	0.000	0.000

Other limits of the study are smaller but real. The Moderate class has seven test participants, so one participant changes its recall by 14 pp. Runs of the same configuration also differ: Experiment 2 and the mean-pooling run of Experiment 3 use the same inputs and differ only in the optimizer, yet their mean macro-F1 values differ by up to 2.3 pp. Some result files contained extra rows from an earlier partial run (two in the BiLSTM file and three in the mean-pooling file of the Experiment 3 self-attention runs); each row was matched against the values printed in the notebook output, the unmatched rows were removed, and the remaining five rows per method reproduce the summary files exactly. Finally, two features of the participant metadata, oxygen saturation and cold hands and feet, were not used, because the column-matching code looked for names without accents and did not find them; the twelve features listed in Section 3.2.6 are those of the executed runs.

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4.3.3.8 What the Ablation Established

Taken together, the ablation supports a cautious set of conclusions. Under the five-fold protocol, six configurations that can be compared fairly produced mean macro-F1 values within about eight percentage points of each other, a range comparable to the fold-to-fold spread. Self-attention fusion was not shown to be better or worse than concatenation, replacing mean pooling by a BiLSTM had no consistent effect, and adding demographic and symptom features did not improve the results where the comparison was fair. Single-split results often disagreed with the five-fold results, which shows that single runs on this test set are unreliable guides to component effects. Finally, the Moderate class was detected rarely or not at all in nearly every configuration, so the macro-F1 values mainly reflect the Normal and Mild classes. The ablation also had a second task, which is to decide which configuration to carry forward, and this is the subject of the next section.

4.3.4 The Selected Model

When one configuration has to be chosen for the next stage, two answers to the question "which model is best?" are possible, because the raw numbers and the trustworthy numbers point to different models. The highest raw scores belong to the BiLSTM model with demographic features, but that result was found unreliable (Section 4.3.3.7) and is excluded from the selection until it is re-run on the common image folders. Among the fairly comparable runs, the image-only model with BiLSTM temporal pooling and self-attention fusion, evaluated with five-fold cross-validation (Ex-1), has the highest mean macro-F1 (0.459 ± 0.075) and the highest fold-averaged macro-F1 (0.468) and accuracy (0.707). It is therefore carried forward as a provisional candidate, and not as a proven winner.

Three reasons support this choice. It leads on the primary metric under both the per-fold and the fold-averaged view. It needs no demographic data, which gave no measurable benefit in the fair comparisons and would complicate a deployment. And it keeps some ability to separate the minority classes, which a screening model needs. The choice of macro-F1 as the priority is a modelling decision and not a clinical one, because the research objective does not state whether missed anemia or false alarms are more costly.

The architecture of the selected model is the one shown in Figure 3.4. It takes the 32 fingertip frames, the 16 palm frames and the nail photograph of a participant, encodes them with three ResNet-18 networks into 256-dimensional embeddings, pools the two video sequences with a BiLSTM of 128 units per direction, and fuses the three modality tokens with one self-attention layer of four heads before a 64-unit classification head produces the three class probabilities. It was trained with class-weighted cross-entropy, Adam (learning rate 10^{-4}), a batch size of 16, dropout of 0.3 and no augmentation, for 10 epochs per fold on 548 participants, with the checkpoint chosen by validation macro-F1.

Its performance is summarized in Tables 4.11 and 4.12. The five fold models differ considerably: their macro-F1 ranges from 0.337 to 0.525, a spread as large as most of the differences between configurations in the ablation. Averaged over the folds, the selected

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model clearly exceeds its concatenation counterpart, which has the same inputs but a macro-F1 of only 0.380 ± 0.041.

Table 4.11: Selected model (image-only, BiLSTM, self-attention): test result of each fold model.

Fold	Accuracy	Macro-P	Macro-R	Macro-F1	Macro-AUC
1	0.683	0.356	0.352	0.337	0.683
2	0.683	0.541	0.517	0.525	0.764
3	0.707	0.592	0.507	0.513	0.720
4	0.675	0.540	0.431	0.451	0.687
5	0.602	0.471	0.494	0.470	0.720

Table 4.12: Selected model compared with its concatenation counterpart (same inputs). Mean ± standard deviation over the five fold models, and the fold-averaged (ensemble) result.

View	Fusion	Acc.	Macro-P	Macro-R	Macro-F1	Macro-AUC
Mean ± SD	Self-att. (selected)	0.670±0.040	0.500±0.092	0.460±0.069	0.459±0.075	0.715±0.033
	Concat.	0.626±0.063	0.384±0.043	0.385±0.042	0.380±0.041	0.675±0.043
Fold-averaged	Self-att. (selected)	0.707	0.562	0.446	0.468	0.756
	Concat.	0.634	0.353	0.360	0.356	0.701

A look at the individual classes shows where the model succeeds and where it does not. The fold-averaged predictions give a precision, recall and F1 of 0.778, 0.885 and 0.828 for Normal (87 participants), of 0.409, 0.310 and 0.353 for Mild (29 participants) and of 0.500, 0.143 and 0.222 for Moderate (7 participants). The confusion matrices of Figure 4.2 make the same point: most Normal participants are recognized, but the Mild participants are largely mistaken for Normal, and the model finds one of the seven Moderate participants where the concatenation model finds none.

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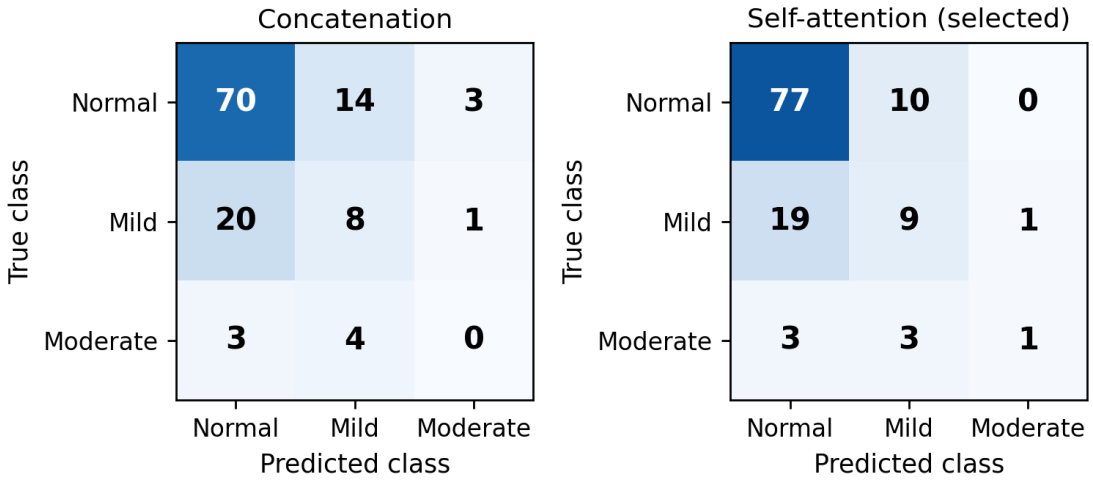


Figure 4.2: Fold-averaged confusion matrices on the test set: concatenation (left) and the selected self-attention model (right).

How convincing is the lead? Table 4.13 places the selected model among the other comparable five-fold configurations. A paired permutation test of its per-fold macro-F1 against concatenation gives a mean difference of +0.080 and $p = 0.178$, which is not significant, and with five folds the smallest attainable p is 0.0625 in any case. Against the simplest alternative, image-only mean pooling with concatenation, the lead is only 1.9 pp, which lies within one standard deviation.

Table 4.13: Selected model among the comparable five-fold configurations.

Configuration	Fusion	Macro-F1	Macro-AUC
Image-only, BiLSTM pooling	**Self-att.**	$\mathbf{0.459 \pm 0.075}$	$\mathbf{0.715 \pm 0.033}$
Image-only, mean pooling	Concat.	0.440 ± 0.089	0.711 ± 0.056
Image-only, mean pooling	Self-att.	0.433 ± 0.040	0.708 ± 0.027
Image + demographics, mean pooling	Self-att.	0.410 ± 0.065	0.692 ± 0.047
Image + demographics, mean pooling	Concat.	0.404 ± 0.053	0.721 ± 0.036
Image-only, BiLSTM pooling	Concat.	0.380 ± 0.041	0.675 ± 0.043

The honest summary is therefore that the selected model is the best of a group of models that are hard to tell apart, and its weaknesses are as clear as its lead. Its fold-averaged accuracy of 0.707 equals that of always predicting Normal, so accuracy offers no support for it. The minority classes remain weak: the Mild F1 is 0.353, and the Moderate class is found in one of seven cases, which rests on a single participant. The result is also not stable across protocols, since the single-split run of the same configuration scored a macro-F1 of 0.406 and a macro-AUC of 0.697, below its concatenation counterpart (0.439). The $\pm$ values describe changes in the training and validation data and not the sampling uncertainty of a 123-participant test set, and in the training histories the validation loss is lowest after the first or second epoch and rises afterwards while the training loss keeps falling. Compared with the tabular logistic regression on the same test set (macro-F1 0.407,

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macro-AUC 0.717), the selected model is ahead by 6.1 pp in fold-averaged macro-F1 and by 3.9 pp in macro-AUC, which is a small margin. No claim is made about computational efficiency, robustness or clinical suitability, because none of these was measured.

4.3.5 Comparative Benchmarking

This subsection asks whether the selected architecture is competitive with methods published elsewhere. The methods, protocol groups and design decisions are defined in Section 3.2.8.

112 The selected model reaches a mean macro-F1 of 0.459 ± 0.075 and a macro-AUC of 0.715 ± 0.033 over five folds. On the fixed test set of 123 participants (87 Normal, 29 Mild, 7 Moderate) its fold-averaged F1 is 0.828 for Normal, 0.353 for Mild and 0.222 for Moderate, where it finds one of seven participants. Its accuracy (0.670) is not informative, because always predicting Normal gives 0.707, so macro-F1 is the primary metric and macro-AUC the secondary one.

4.3.5.1 Overall Scores

Table 4.14 gives the test results of the selected model and of one run per paper, and Figure 4.3 shows them as bars.

Table 4.14: Test results of the selected model and of one run per paper. Δ = benchmark − selected model. † Single run on its own test split. ‡ Mean ± SD over five participant-level folds of all nail participants. The other rows are the mean ± SD over five folds on the fixed test set.

ID	Method	Acc	Macro-F1	Macro-AUC	ΔF1	ΔAUC
Sel.	**Selected model (BiLSTM + self-attention)**	**0.670**	**0.459 ± 0.075**	**0.715 ± 0.033**	—	—
B03	EfficientNet-B4 + a^* colour	0.550	0.361 ± 0.043	0.602 ± 0.052	−0.099	−0.113
B04	Intermediate fusion (MobileNetV3-S)	0.541	0.328 ± 0.022	0.591 ± 0.017	−0.132	−0.124
B05	AFF (ResNet-18)	0.572	0.405 ± 0.035	0.702 ± 0.037	−0.055	−0.013
B06	Late ElasticNet average	0.533	0.419 ± 0.020	0.701 ± 0.015	−0.040	−0.014
B07	k-NN ($k = 5$), colour features	0.685	0.410 ± 0.054	0.613 ± 0.022	−0.050	−0.102
B08	Majority vote, 3 × ResNet-18	0.620	0.418 ± 0.037	0.694 ± 0.015	−0.042	−0.021
B09	BPANet-adapted†	0.691	0.416	0.762	−0.043	+0.047
B10	MV-HFMD†	0.667	0.267	0.607	−0.193	−0.108
B11	Nail-only DenseNet169†	0.500	0.417	0.663	−0.043	−0.052
B12	RCBAM-MobileNetV2, image only†	0.665	0.266	0.577	−0.193	−0.138
B13	EfficientNet-B0 + GeM‡	0.578	0.389 ± 0.062	0.645 ± 0.066	−0.070	−0.070

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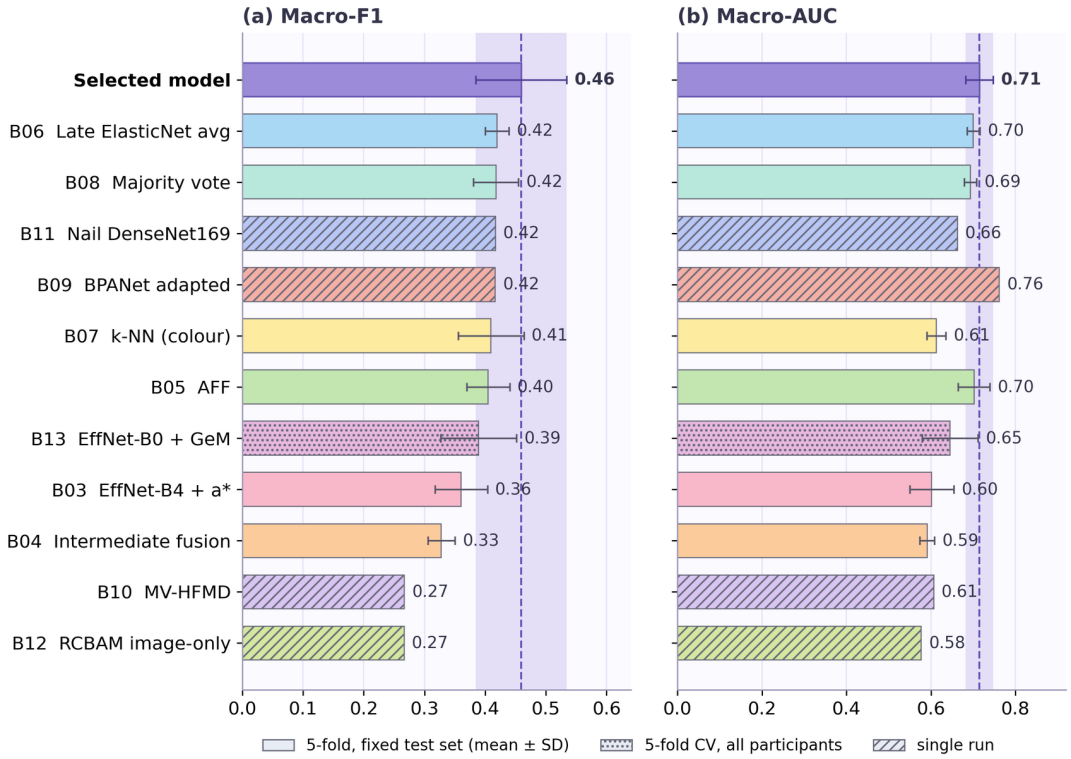


Figure 4.3: Macro-F1 (a) and macro-AUC (b), one run per paper. Solid bars: five-fold mean ± SD on the fixed test set. Dotted bar: five-fold CV over all participants. Hatched bars: single runs. The dashed line and band show the mean ± SD of the selected model.

Macro-F1 and AUC. The selected model has a higher macro-F1 than the run reported for every paper. The differences fall into two groups. Seven methods lie within one standard deviation (0.075) of the selected model: AFF, late ElasticNet average, k-NN, majority vote, BPANet-adapted, the nail-only DenseNet and the nail EfficientNet-B0 (macro-F1 0.389 to 0.419). With the available evidence they cannot be separated from the selected model. Four are clearly lower: B03 (0.361), B04 (0.328) and, by the widest margin, MV-HFMD and the RCBAM image branch (both 0.267, about 2.6 standard deviations below). The last two predict Normal for every test participant. On AUC, the only run above the selected model is the single-run BPANet-adapted model (0.762); one run cannot show that it ranks participants better. Among the five-fold benchmarks on the fixed test set, no paired permutation test reached $p < 0.0625$, the smallest value attainable with five folds, so no difference is statistically established.

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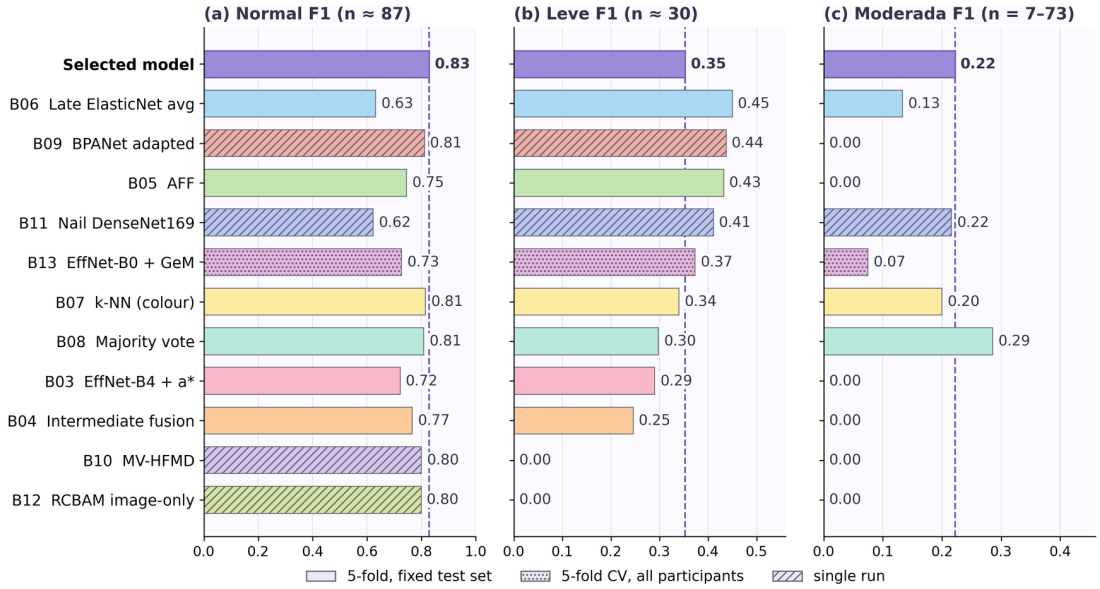


Figure 4.4: Per-class F1 for (a) Normal ($n \approx 87$), (b) Mild ($n \approx 30$) and (c) Moderate ($n = 7$ to 73, depending on the test set). The figures label the classes *Leve* (Mild) and *Moderada* (Moderate). Hatched and dotted bars as in Figure 4.3. The dashed line marks the selected model.

Normal class. The selected model has the highest Normal F1 (0.83), but k-NN, BPANet-adapted and majority vote reach 0.81, and even the two runs that predict Normal for everyone reach 0.80 (Figure 4.4a). Normal F1 therefore says little about whether anemia is detected.

Minority classes. The methods differ on the two small classes (Figures 4.4b and 4.4c). Five methods have a higher Mild F1 than the selected model (0.353): late ElasticNet average (0.450), BPANet-adapted (0.438), AFF (0.432), the nail-only DenseNet (0.411) and the nail EfficientNet-B0 (0.373). On Moderate, six methods score 0.000. Only majority voting (0.286) exceeds the selected model (0.222), and the nail-only DenseNet is level with it (0.216). With seven to eleven Moderate participants in most test sets, a single correct prediction moves recall by 9 to 14 percentage points, so no ranking on this class is reliable.

4.3.5.2 Class-wise Errors and Screening View

Macro-F1 does not show where the errors of a model fall. Table 4.15 gives the recall of each class and a Normal-versus-anemic screening view obtained by merging Mild and Moderate (the models were not trained for this binary task), and Figures 4.5 and 4.6 show the confusion matrices. The selected model and B03 to B08 are scored on the same 123 participants, using the averaged predictions of the five fold models, so these rows can be compared with one another; B09 to B13 are scored on their own splits, so their rows describe each run and do not rank it.

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Table 4.15: Class-wise recall (with counts) and Normal-versus-anemic screening (sensitivity and specificity, with counts and 95% Wilson intervals). Selected model and B03 to B08: averaged predictions of the five fold models on the same 123 test participants. † Single run on its own test split. ‡ Predictions pooled over the five folds.

Run	Normal recall	Mild recall	Moderate recall	Sensitivity	Specificity
Selected model	0.885 (77/87)	0.310 (9/29)	0.143 (1/7)	0.39 (14/36) [0.25, 0.55]	0.89 (77/87) [0.80, 0.94]
B03 EffNet-B4 + a^*	0.701 (61/87)	0.310 (9/29)	0.000 (0/7)	0.42 (15/36) [0.27, 0.58]	0.70 (61/87) [0.60, 0.79]
B04 Intermediate fusion	0.793 (69/87)	0.241 (7/29)	0.000 (0/7)	0.33 (12/36) [0.20, 0.50]	0.79 (69/87) [0.70, 0.86]
B05 AFF	0.690 (60/87)	0.552 (16/29)	0.000 (0/7)	0.61 (22/36) [0.45, 0.75]	0.69 (60/87) [0.59, 0.78]
B06 Late ElasticNet avg.	0.494 (43/87)	0.621 (18/29)	0.286 (2/7)	0.83 (30/36) [0.68, 0.92]	0.49 (43/87) [0.39, 0.60]
B07 k-NN (colour)	0.885 (77/87)	0.276 (8/29)	0.143 (1/7)	0.31 (11/36) [0.18, 0.47]	0.89 (77/87) [0.80, 0.94]
B08 Majority vote	0.828 (72/87)	0.241 (7/29)	0.429 (3/7)	0.47 (17/36) [0.32, 0.63]	0.83 (72/87) [0.73, 0.89]
B09 BPANet adapted†	0.816 (71/87)	0.483 (14/29)	0.000 (0/7)	0.53 (19/36) [0.37, 0.68]	0.82 (71/87) [0.72, 0.88]
B10 MV-HFMD†	1.000 (82/82)	0.000 (0/30)	0.000 (0/11)	0.00 (0/41) [0.00, 0.09]	1.00 (82/82) [0.96, 1.00]
B11 Nail DenseNet169†	0.518 (43/83)	0.500 (15/30)	0.364 (4/11)	0.71 (29/41) [0.56, 0.82]	0.52 (43/83) [0.41, 0.62]
B12 RCBAM image-only†	1.000 (165/165)	0.000 (0/61)	0.000 (0/22)	0.00 (0/83) [0.00, 0.04]	1.00 (165/165) [0.98, 1.00]
B13 EffNet-B0 + GeM‡	0.702 (386/550)	0.420 (84/200)	0.068 (5/73)	0.54 (147/273) [0.48, 0.60]	0.70 (386/550) [0.66, 0.74]

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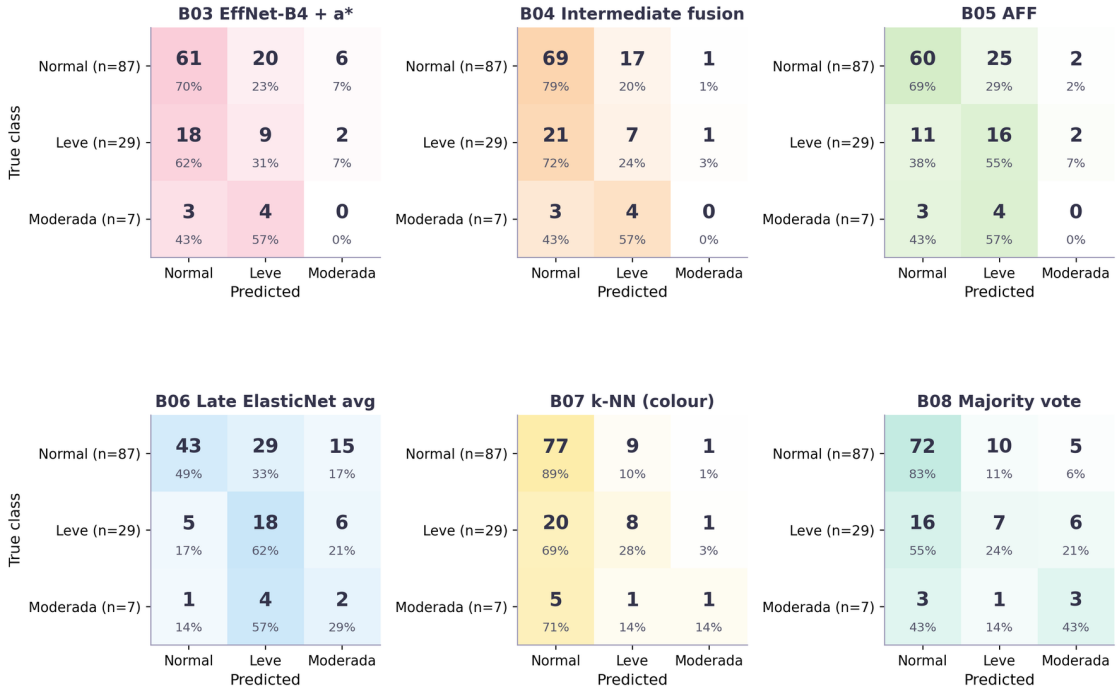


Figure 4.5: Confusion matrices of B03 to B08 on the common test set of 123 participants (averaged predictions of the five fold models). Large numbers are counts; the percentage is the share of each true class. The figure labels the classes *Leve* (Mild) and *Moderada* (Moderate).

Runs on the common test set. The selected model misses 22 of 36 anemic participants (sensitivity 0.39) but wrongly flags only 10 of 87 Normal participants (specificity 0.89). B07 has almost the same profile, and B04 and B08 lie between it and the others. B03 and B05 flag more anemic participants (0.42 and 0.61) at the cost of specificity (0.70 and 0.69). B06 is the extreme case: it finds 30 of 36 anemic participants but only 43 of 87 Normal participants are correct, and 15 Normal participants are called Moderate; both of its intervals are clearly separated from those of the selected model. On Moderate, B03, B04 and B05 find none of the seven participants, the selected model and B07 find one, B06 two and B08 three. The macro-F1 values in Table 4.14 are close because the methods trade sensitivity against specificity at different points, not because they make the same errors.

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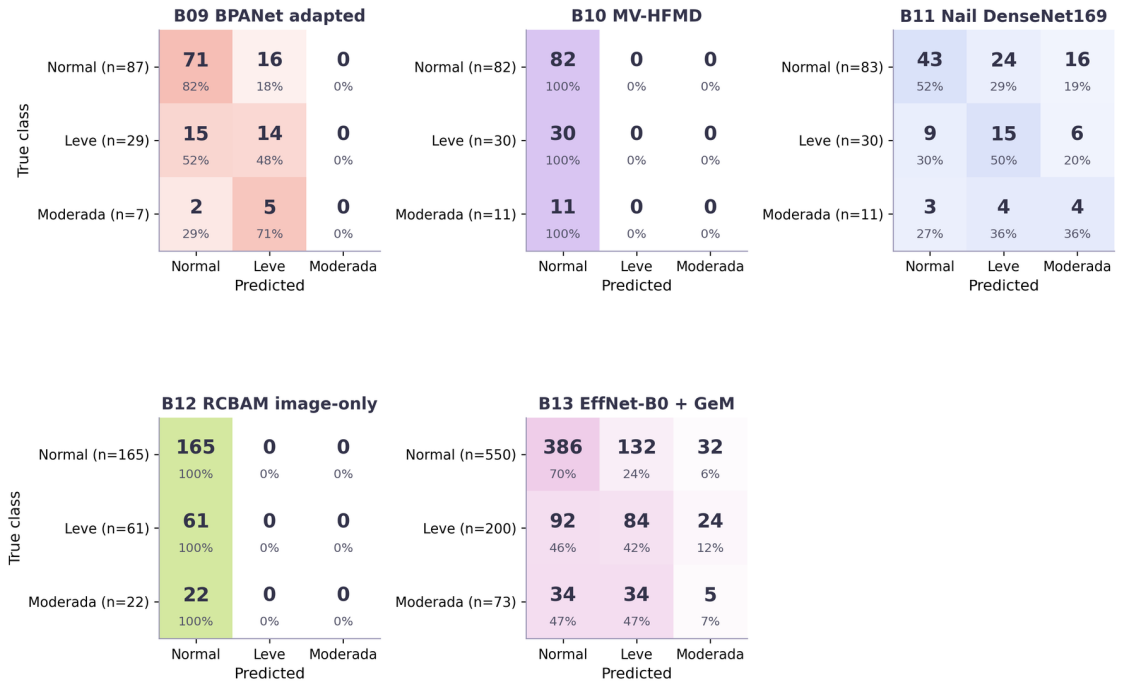


Figure 4.6: Confusion matrices of the single runs B09 to B12 and of the cross-validated B13, each on its own test set. Large numbers are counts; the percentage is the share of each true class. The figure labels the classes *Leve* (Mild) and *Moderada* (Moderate).

Runs on their own splits. BPANet-adapted finds 14 of 29 Mild participants but none of the seven Moderate ones. The nail-only DenseNet is the only single run that finds Moderate participants at a useful rate (4 of 11), but only 43 of 83 Normal participants are correct. B13 is more moderate, with 5 of 73 Moderate participants found. MV-HFMD and the RCBAM image branch predict Normal for everyone. The selected model is conservative, whereas the nail-only DenseNet (sensitivity 0.71, specificity 0.52) and B06 are far more aggressive; the intervals overlap for most pairs, so no operating point is confirmed as better.

What the training logs explain. The MV-HFMD checkpoint was chosen by validation accuracy and falls on epoch 2, where its validation macro-F1 (0.267) equals that of an all-Normal predictor. BPANet-adapted stayed at the all-Normal level for five of its first six epochs and peaked at a validation macro-F1 of 0.328, yet scored 0.416 on the test set, which shows how noisy single runs are on this dataset.

4.3.5.3 Interpretation

The selected model keeps its place, but its lead is small. No run exceeds it on macro-F1, but seven of the eleven are within one standard deviation, and the order of the

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selected model and BPANet-adapted flips if the selected model misses its single correct Moderate participant. The evidence is best read as a tie with a small edge.

The methods differ mainly in their operating point. The selected model has the highest macro-precision (0.500) and the highest Normal F1 and is conservative, whereas the late ElasticNet average has a higher macro-recall (0.484 against 0.460) and five methods have a higher Mild F1. On the common test set the sensitivity ranges from 0.31 (B07) to 0.83 (B06) and the specificity from 0.49 to 0.89 (Table 4.15).

Nail-only models work only when the training is sound. The pretrained, augmented, class-weighted nail models reach a macro-F1 of 0.417 (B11) and 0.389 (B13), which is 91% and 85% of the selected model, whereas the unweighted B12 (0.266) predicts Normal for everyone. This suggests that the nail photograph carries a usable part of the signal and that the training recipe matters more than the architecture, but the splits and budgets differ and no nail-only baseline was run under the common protocol, so the claim is weak.

Poor results reflect the adaptation, not necessarily the method. MV-HFMD fails because of its checkpoint rule on an imbalanced validation set, B12 because it has no class weighting, and B03 and B04 are three-class re-implementations of originally binary papers. Published accuracies cannot be ranked against these values and give no support for placing any paper above or below the selected model.

4.3.5.4 Limitations and Summary of the Benchmark

The benchmark is not a perfectly fair contest. It uses three protocols, the single runs have no measured run-to-run variation, and B13 is not scored on the test participants of the selected model. The selected model was chosen among several configurations on the same test set, which may favour it, and some variants that were evaluated are not reported: naive Bayes on the colour features of B07 (0.480 ± 0.017, $p = 0.6875$) scored above the mean of the selected model, so the selected model is not the highest of all 17 variants of the first member, although this variant is not the headline method of its paper and the difference is not significant. For B13, the fingertip EfficientNet-B0 of the third member reached 0.447 under five-fold cross-validation, below the selected model. Moderate has only 7 to 73 test participants.

In summary, none of the eleven runs exceeded the selected model on macro-F1; seven lie within one standard deviation of it, four are clearly lower, and BPANet-adapted exceeds it only on AUC and Mild F1. A ranking would need the same split for every method, several seeds or repeated cross-validation, and more Moderate participants.

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4.3.6 Discussion

The results are now brought together to answer the research questions and to relate them to the original problem.

RQ1: what each modality can do alone. Under a basic recipe, each of the nail, palm and fingertip modalities carried a detectable but modest signal, with accuracy of 56 to 66% and AUC-ROC of 0.62 to 0.67. After the ablation, the best unimodal models reached 67.2% (palm), 67.3% (nail) and 73.9% (fingertip), with AUC-ROC of 0.69, 0.71 and 0.74. For the nail, this is close to the accuracy of about 0.70 and the AUC of about 0.74 that have been reported for a DenseNet on the same population[13], which suggests that a small ResNet-18 does not lose much on this modality, although the data partitions are not identical. None of the unimodal models reached the 80% accuracy target of Section 3.1, and their sensitivities of 61 to 67% are insufficient for a screening tool whose purpose is to avoid missing anemic patients.

RQ2: sampling and training strategy. The fingertip modality, whose signal is temporal, responded most to denser sampling and augmentation, while the palm responded least. This pattern is plausible, because the palm video mainly records a hand that opens and its frames are visually repetitive, but the experiments were not designed to test such explanations, and they are offered as hypotheses. Because the best of several configurations was reported on test sets of limited size, the apparent gains are partly subject to selection and should not be regarded as precise estimates.

RQ3: fusion and temporal aggregation. Fusion results can only be compared among themselves, because the fusion experiments use a different label space and, for the binary reference, a different cohort and test set from the single-modality models. Among the reliable five-fold results, self-attention was not shown to be better or worse than concatenation (mean macro-F1 differences of -0.7, $+0.6$ and $+8.0$ pp, no p-value below 0.178), and the BiLSTM had opposite effects for the two fusion modules (-6.1 and $+2.6$ pp). The configuration with the highest macro-F1 combines the two more complex components (BiLSTM and self-attention), but its lead over the simplest configuration, mean pooling with concatenation, is 1.9 pp and lies within one fold standard deviation. A plausible reason for the weak effect of the BiLSTM is that the frames of a fixed body part are visually similar, so that a recurrent model adds parameters without much sequential structure to exploit; this was not tested. The title's emphasis on attention-based fusion should therefore be read as a description of the selected model and not as a demonstrated advantage of attention.

RQ4: demographic and symptom variables. The twelve features alone were weak (macro-F1 0.407 for logistic regression and 0.344 for the random forest), and adding them to the images lowered the mean macro-F1 by 2 to 4 pp under mean pooling, within the

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fold spread. No benefit was therefore shown, and the images carry the discriminative signal that was measured. A relevant confounder is the sex distribution of the cohort: 46.9% of the women but only 9.2% of the men are anemic, so sex is predictive of the label, and an image model could in principle exploit sex-related appearance instead of pallor. This possibility was not examined. It should also be noted that the strongest result with demographic features is unreliable (Section 4.3.3.7) and that two features of the metadata were not used.

RQ5: comparison with other model frameworks. None of the eleven re-implemented methods exceeded the selected model on macro-F1, but seven lay within one standard deviation and no test reached $p < 0.0625$ (Table 4.14). The methods differ mainly in operating point: the selected model is conservative (sensitivity 0.39, specificity 0.89), while the late ElasticNet average finds 30 of 36 anemic participants at a specificity of 0.49 (Table 4.15). The result is a tie with a small edge, in line with the fusion ablation, and the single runs describe their methods without ranking them.

Relation to the original problem. The thesis set out to establish whether smartphone recordings of three body sites can support non-invasive anemia screening and how they should be combined. The evidence indicates that the recordings contain a signal that is detectable with a small pretrained network: single modalities reach 67 to 74% binary accuracy under the best configurations, and the three-class fusion models reach a fold-averaged AUC of about 0.70 to 0.76. The fusion models are, however, not clearly better than a logistic regression on twelve tabular features (macro-AUC 0.717), and no design choice among fusion module, pooling and demographics shows a consistent effect. The evidence does not establish clinical readiness: the performance on the minority classes is poor, the evaluation relies on a small single-site test set with seven Moderate participants, and the limitations described next affect how far the numbers can be trusted.

4.3.7 Limitations of the Evidence

Inconsistency in the binary label file. After the experiments, the class and binary-label columns of the processing table `Dataset2.csv` were compared with the participant metadata file. For the participants with identifiers ID058 to ID227 the class entries of the table equal those of the metadata shifted by one row, so that 72 of the 909 binary labels (7.9%) disagree with the metadata, 36 in each direction; no disagreement was found outside this range. The class folder recorded in the file paths of the table follows the shifted class, so frames organized by class folder carry the same error. For the 82 participants whose three-class entry differs, the metadata class is the one closer to the recorded hemoglobin value in 71 cases, which supports the metadata as the correct source. Among the 125 test participants with all three recordings in the split defined by the table, 12 labels (9.6%) are affected. The binary single-modality experiments and Experiment 0 used labels from this table, so their results were obtained with this label noise and should be interpreted

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with caution; Experiment 0 also had 737 usable participants instead of 808, which fits the shifted class column causing some file look-ups to fail, although the notebooks do not confirm this. The three-class experiments used the metadata labels and are not affected by this specific problem.

Unreliable BiLSTM run with demographic features. The five-fold results of Experiment 3 with BiLSTM pooling (macro-F1 0.668 and 0.687) read their images from different input folders and classify all Mild participants correctly, unlike every other configuration (Section 4.3.3.7). They are excluded from all conclusions until the experiment is repeated on the common image folders and the stored images are compared class by class.

Design confounds and weak statistics. The fusion comparison changes the head size and the fused-vector size together with the fusion rule. Every cell of the single-split tables is one run with one seed, the only statistical test (five-fold sign-flip permutation) cannot reach $p < 0.05$, and there are no confidence intervals, bootstrap or McNemar tests because per-participant predictions were not saved. Many configurations were compared on one test set and the best stands out, which is a risk of optimistic selection, although each run used the test set only once after the checkpoint was chosen. Parameter counts, training time and single-modality ablations (fingertip only, palm only, nail only) were not measured. The comparative benchmarking against other model frameworks is reported separately in Section 4.3.5.

Small and imbalanced test sets. The Moderate class contains seven test participants, so one participant changes its recall by 14 percentage points. The 95% confidence interval of an accuracy near 70% on a test set of 110 to 126 participants is about ±8 to 9 percentage points. No repeated runs with different seeds were performed outside the five-fold cross-validation, and the five-fold standard deviation does not capture the sampling uncertainty of the test set.

Selection effects and differing protocols. Only the best configuration of each modality is reported, and the test sets, label spaces and evaluation units (frame or participant) differ between groups of experiments. Gains over the baseline are therefore optimistic estimates, and cross-group comparisons are indicative only.

Benchmarking protocols. Only the benchmark runs B03 to B08 can be compared with the selected model directly; the single runs have no run-to-run variation, B13 includes the test participants of the selected model in its folds, and the selected model may be favoured by having been chosen on the same test set (Section 4.3.5.4).

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Reference standard and generalization. Hemoglobin was measured non-invasively, and the class thresholds were set locally. All data come from one site, one phone model and participants aged 18 to 25, with no severe cases, so generalization to children, older adults, other devices or other populations is not supported.

Computational resources. The use of free-tier GPUs limited hyperparameter tuning and the number of repeated runs.

4.4 Summary

The single-modality baselines reached 56 to 66% accuracy, and the ablation raised the best fingertip, nail and palm models to 73.9%, 67.3% and 67.2%, with the fingertip model gaining most from dense sampling and augmentation. The fusion ablation, run under single-split and five-fold protocols on a common three-class test set, found no consistent effect of the fusion module, the temporal pooling or the demographic features; fairly comparable configurations lay between 0.380 and 0.459 macro-F1, close to a tabular logistic regression (0.407). The image-only model with BiLSTM pooling and self-attention fusion (five-fold macro-F1 0.459 ± 0.075, fold-averaged macro-AUC 0.756) was selected as a provisional candidate, but it recognized only one of seven Moderate participants. The comparative benchmarking of eleven re-implemented published methods found none that exceeded it on macro-F1, but seven lay within one standard deviation, so the result is a tie with a small edge. A label inconsistency in the binary label file, a small test set and an unreliable BiLSTM run with demographic features limit how far these results can be trusted. Chapter 5 documents the standards and constraints of the work, and Chapter 6 draws the conclusions.

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Chapter 5

Standards and Design Constraints

The preceding chapters described what was built and what it achieved. This chapter documents the conditions under which the work was carried out: the standards followed, the constraints that shaped the design, the cost of the study, and the mapping of the project to the complex engineering problem and activity attributes of the degree programme. These conditions explain several choices made earlier, such as the use of a compact pretrained network and free cloud computing.

5.1 Compliance with the Standards

Only the standards that relate to this research project are discussed. For each, the available alternatives are compared with their advantages and disadvantages, and the reason for the choice is given.

5.1.1 Software Standards

The software stack consists of Python 3.12 (3.12.13 in the final experiments) with PyTorch for the models, Pandas and NumPy for data handling, Scikit-learn for metrics, cross-validation splits and the tabular baselines, and OpenCV for video and image preparation. All frames and images were resized to 224×224 pixels and normalized with the ImageNet statistics, as required by the pretrained ResNet-18; the final layer has one output for the binary task and three for the three-class task. GitHub and the Colab and Kaggle notebook histories were used for version control, and the metrics of every run were stored in CSV files for comparison. Table 5.1 compares the choices with their alternatives.

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Table 5.1: Software choices, alternatives and rationale.

Choice	Alternatives	Advantages and disadvantages	Rationale
PyTorch	TensorFlow with Keras; JAX	PyTorch: flexible dynamic graphs, easy debugging, large collection of pretrained vision models; deployment to phones needs a conversion step. TensorFlow with Keras: mature mobile and server deployment tooling, but less common in current research code. JAX: fast and functional, but a smaller ecosystem of pretrained vision models and a steeper learning curve.	Direct access to the ImageNet-pretrained ResNet-18 and easy implementation of custom fusion modules.
Python 3.12	MATLAB; R	Python: broad machine learning ecosystem and free of charge. MATLAB: strong toolboxes but licensed. R: strong statistics, weaker deep learning support.	Free, and all required libraries are available.
OpenCV	Pillow; torchvision video readers	OpenCV: reads video, converts colour spaces (the YCrCb space was used to locate the nail) and crops. Pillow: still images only. torchvision readers: fewer image-processing functions.	One library covers frame extraction, nail localization and resizing.
Scikit-learn	Custom code; XGBoost	Scikit-learn: consistent interface for metrics, folds and the logistic regression and random forest baselines. XGBoost: strong on tabular data but an extra dependency, and its integration was dropped from the plan (Section 3.3).	The baselines only need standard classifiers and metrics.

5.1.2 Hardware Standards

All experiments ran on free-tier cloud platforms (Google Colab and Kaggle Notebooks) with NVIDIA Tesla T4 GPUs, a P100 in some Colab sessions, and two T4 GPUs on Kaggle for the fusion experiments. The detailed specification is given in Section 4.1. Table 5.2 compares this choice with its alternatives.

Table 5.2: Hardware choice, alternatives and rationale.

Option	Advantages and disadvantages	Rationale
Free-tier cloud GPUs (chosen)	No cost and reproducible by others; session length, memory (15–16 GB of GPU memory) and GPU type are limited and vary between sessions.	Sufficient for ResNet-18 models and consistent with the economic constraint.
Local GPU workstation	No session limits and persistent storage; purchase and maintenance cost.	Not used in this project.
Paid cloud GPUs	Longer sessions, more memory and faster training; recurring cost (Section 5.3).	Considered for larger-scale development only.

5.1.3 Communication Standards

The study did not build a communication system. The dataset was obtained from a public repository, and the experiments ran inside the cloud notebooks. A deployed screening platform, such as the one sketched in Figure 3.1, would additionally need secure transport of recordings and results (TLS) and an agreed format for exchanging health data. A

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health-data standard such as HL7 FHIR would allow results to be exchanged with health records and is more interoperable than a custom REST interface, but it is more complex to implement; a custom interface would be quicker to build but harder to integrate. This choice is left to future development.

5.1.4 Data Standards

The original data are organized in three folders for the modalities, fingertips (*yemas*), nails (*uñas*) and palms (*palmas*), each with class subfolders. The nail photographs are JPG files and the fingertip and palm recordings are MP4 files; extracted frames were stored as JPG. The dataset classes are Normal, Mild (*Leve*) and Moderate (*Moderada*), and the binary label merges the last two. Participant names are not present, and records are linked by anonymous identifiers (for example ID001). The dataset is distributed under CC BY 4.0 and is cited accordingly.

Storing frames as JPG keeps the storage need within the free Google Drive tier, whereas a lossless format such as PNG would avoid compression artefacts but multiply the storage need. Whether JPG compression affects the subtle colour information that the task depends on was not studied.

5.2 Design Constraints

The study was designed for a low-cost, non-invasive setting and for cloud-based computing, with attention to the medical sensitivity of the task.

5.2.1 Economic Constraint

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All experiments were run on free-tier cloud resources, so that other researchers can reproduce and extend the work without specialized hardware. The consequence is limited session length and memory, which restricted hyperparameter tuning.

5.2.2 Environmental Constraint

The screening approach under study requires no consumables such as needles, tubes or reagents, which reduces medical waste compared with blood sampling. The environmental benefit is a property of the proposed approach, not a measured outcome of this study, and the computation itself consumes energy in cloud data centers.

5.2.3 Ethical Constraint

The data are anonymized in the original release, and only anonymous identifiers were used. Because a screening model can produce false negatives, results should not be used for clinical decisions, and any deployment would require ethical approval and clinical validation.

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5.2.4 Health and Safety Constraint

In screening, a false negative leaves an anemic person unidentified. The single-modality models detect about 55% to 67% of the anemic cases and the binary fusion reference model 20% to 48% (single split and five-fold, Chapter 4), while the selected three-class model finds one of seven Moderate participants, so none of them yet satisfies the safety expectations of a stand-alone screening tool. Reducing false negatives is therefore the main priority for further development.

5.2.5 Social Constraint

A photograph- or video-based method avoids the discomfort and the small infection risk of blood draws and could lower the cost of screening in rural areas where laboratory testing is expensive. Acceptance in the target communities has not been studied.

5.2.6 Political Constraint

A screening model for a medical condition falls under the regulation of medical software and under health-data protection law in any country in which it would be deployed. Neither was studied here, because the models were developed and evaluated as research prototypes on a public, anonymized dataset. Any deployment would require the approvals of the relevant health authority.

5.2.7 Sustainability

ResNet-18 is a lightweight pretrained network that can be maintained and retrained without high-performance hardware, which supports the long-term use of the approach.

5.2.8 Technical Constraint

The method depends on recordings of the fingertip, the palm and the nail that are acquired under conditions similar to those of the dataset (fixed phone, controlled lighting for the nail photograph, defined hand movements). Preprocessing normalizes the inputs, but the performance under other conditions is unknown. The recordings were kept on Google Drive, which offers 15 GB in its free tier, and frames were extracted to temporary cloud storage; extracting 8, 16 or 32 frames per video multiplies the storage need accordingly. The output of the model is a simple class and its probability, so that users do not need programming knowledge to read it, but a usable interface was not built in this study.

5.3 Cost Analysis

This section lists the cost of the study and an estimate of the cost of a larger development effort.

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5.3.1 Actual Budget

Item	Specification	Cost
Computing hardware	Kaggle or Google Colab (free tier), NVIDIA T4 GPU	$0.00 (0 BDT)
Development software	Python, PyTorch, Matplotlib (open source)	$0.00 (0 BDT)
Dataset	Public fingertip, palm and nail dataset (CC BY 4.0)	$0.00 (0 BDT)
Cloud storage	Google Drive (15 GB free tier)	$0.00 (0 BDT)
Total		**$0.00 (0 BDT)**

Table 5.3: Actual budget of the research study.

Table 5.3 shows that the study used no paid resources; the cost excludes the time of the authors and supervisors.

5.3.2 Alternative Budget for Large-Scale Development

The limits of the free tier suggest what a larger study would need. Table 5.4 gives an estimate for a scaled-up development.

Item	Estimated cost	Rationale
Google Colab Pro+	$50 (6112.87 BDT) / month	Hyperparameter tuning and repeated cross-validation.
Google One	$10 (1222.57 BDT) / month	Storage of high-quality recordings in the cloud.
Field research	$200 (24451.5 BDT) / month	Collection of clinical data to compare predictions with laboratory results and to assess safety.
Total	**$260 (31786.95 BDT) / month**	**Large-scale development.**

Table 5.4: Estimated budget for large-scale development.

5.3.3 Revenue Model

The study itself produces no revenue, and the revenue model below describes how a later development could be financed. Three sources are conceivable. First, the method could be licensed to developers of health applications, who would pay a licence or integration fee. Second, non-governmental health organizations and public health programmes that run field screening could pay for the tool through a subscription or a fee per screening. Third, research grants from health foundations could finance validation against laboratory hemoglobin, which any commercial use would require first. These are proposals only: none has been pursued, no prices have been tested, and the alternative budget of Table 5.4 would have to be covered by the first source or by grants.

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5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

Table 5.5 maps the project to the attributes of a complex engineering problem.

P1	P2	P3	P4	P5	P6	P7
Depth of knowledge	Range of conflicting require-ments	Depth of analysis	Familiarity of issues	Extent of applicable codes	Extent of stakeholder involve-ment	Inter-dependence
✓		✓	✓			

Table 5.5: Mapping with complex problem solving.

5.4.1.1 P1: Depth of knowledge

The work requires knowledge of convolutional networks, transfer learning, temporal modeling of video, multimodal fusion and the evaluation of classifiers on imbalanced medical data.

5.4.1.2 P1: Knowledge profile mapping

Table 5.6 maps the depth of knowledge required by P1 to the knowledge profile attributes.

ID	**Knowledge profile attribute**	**Mapped**
K1	Natural sciences	
K2	Mathematics	
K3	Engineering fundamentals	✓
K4	Engineering specialist knowledge	✓
K5	Engineering design	✓
K6	Engineering practice	
K7	Comprehension (role of engineering in society)	✓
K8	Research literature	✓

Table 5.6: Mapping of P1 with the knowledge profile.

K3: Engineering fundamentals. The project rests on the fundamentals of computing and data processing: image and video representation, optimization of neural networks and the evaluation of classifiers.

K4: Specialist knowledge. The models require specialist knowledge of convolutional networks, transfer learning, temporal modelling of video and multimodal fusion with self-attention (Chapter 2).

K5: Engineering design. The fusion framework and the ablation studies were designed to answer the research questions under limited data and computing resources (Chapter 3).

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K7: Comprehension. The project required an understanding of the role of screening in health care, of the consequences of false negatives and of the ethical limits on using such a model (Section 5.2).

K8: Research literature. The design builds on the literature on smartphone-based and image-based anemia detection and on multimodal fusion, and the gap that it addresses was derived from that literature (Section 2.3).

5.4.1.3 P3: Depth of analysis

The task requires the analysis of subtle visual differences in noisy recordings with few labeled examples and a very small minority class, and the careful interpretation of results obtained on small test sets.

5.4.1.4 P4: Familiarity of issues

Anemia is conventionally diagnosed with a blood test. Estimating hemoglobin-related status from smartphone recordings is an unconventional approach for which standard solutions and validation procedures are not yet established.

5.4.2 Engineering Activities

Table 5.7 maps the project to the attributes of complex engineering activities.

A1	A2	A3	A4	A5
Range of resources	Level of interaction	Innovation	Consequences for society and environment	Familiarity
✓		✓	✓	

Table 5.7: Mapping with complex engineering activities.

5.4.2.1 A1: Range of resources

The project combined a public multimodal dataset of video and images, pretrained deep networks, a deep learning framework (PyTorch), libraries for evaluation, and cloud GPUs.

5.4.2.2 A3: Innovation

The project investigates a non-invasive alternative to blood sampling and compares fusion designs for three visual modalities through an ablation study, an application of deep learning to a medical problem for which no established solution exists.

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5.4.2.3 A4: Consequences for society and environment

If validated, a non-invasive screening method could improve access to anemia screening for underserved communities and avoid the waste associated with blood sampling. The consequences of false negatives, discussed in Section 5.2, are a reason for caution.

5.5 Summary

This chapter set out the standards, constraints and costs of the study. The use of free cloud resources and a compact network made the work reproducible at no cost but limited tuning and repeated runs, and the safety constraint shows that the current models are not suitable for stand-alone clinical use. Chapter 6 concludes the thesis by returning to the research problem.

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Chapter 6

Conclusion

This thesis began with a practical problem: anemia is common, its standard diagnosis requires a blood sample, and a screening method that needs only a smartphone camera could extend testing to settings where laboratories are scarce. The research question was whether recordings of the fingertip, the palm and the fingernail can support such screening when analyzed with deep learning, and how the three sources should be combined. This chapter summarizes what the work established, states the limitations that qualify it, and outlines the work that would resolve what remains open.

6.1 Summary

A public dataset of 909 young adults, with fingertip videos, palm videos, nail photographs, demographic data and hemoglobin-based classes, was prepared with patient-level stratified splits. Single-modality ResNet-18 baselines reached accuracies of 56% to 66%. An ablation of sampling density and training strategy then showed that the fingertip video benefited most from 32 frames and augmentation (accuracy 73.87%, AUC-ROC 0.743), that the nail photographs improved to 67.27% accuracy and an AUC-ROC of 0.712 with weighted loss and augmentation, and that the palm video improved marginally to 67.20%.

A tri-modal fusion framework with ResNet-18 encoders was then studied through a component-wise ablation (fusion module, temporal pooling, demographic and symptom variables) on a common three-class test set of 123 participants, under single-split and five-fold protocols. Self-attention was not shown to be better or worse than concatenation, the bidirectional LSTM had no consistent effect compared with mean pooling, and demographic and symptom variables, which are weak predictors alone, did not improve the image-based models where the comparison was fair. Six fairly comparable configurations reached a mean five-fold macro-F1 of 0.380 to 0.459, close to the 0.407 of a logistic regression on the tabular variables. The image-only model with BiLSTM pooling and self-attention fusion had the highest macro-F1 (0.459 ± 0.075; fold-averaged macro-F1 0.468 and macro-AUC 0.756) and is reported as a provisional candidate, although it identified only one of the seven Moderate participants and its lead is within one fold standard devi-

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ation.

Research questions and contributions. The thesis answers the five research questions as follows. For RQ1, each single modality carries a detectable but modest signal under a basic recipe. For RQ2, denser sampling and augmentation matter most for the fingertip video and little for the palm video. For RQ3, neither self-attention fusion nor BiLSTM pooling showed a consistent advantage over concatenation and mean pooling; the highest macro-F1 was obtained by combining the two complex components, but the lead is within the fold-to-fold spread. For RQ4, demographic and symptom variables are not sufficient alone and show no benefit as an addition to the images in the fair comparisons. For RQ5, none of the eleven re-implemented published methods exceeded the selected model on macro-F1, but seven lay within one standard deviation of it, so the result is a tie with a small edge and not a demonstrated advantage. The contributions are the documented preprocessing and evaluation pipeline, the ablation of frame sampling and training strategy, the fusion framework with its component-wise ablation and comparison with tabular baselines, the selection and description of a provisional best model, the comparative benchmarking against eleven re-implemented published methods, and an explicit account of the limitations of the evidence.

Concluding remarks. The thesis started from the need for a screening method that does not depend on a blood sample, and it has shown that fingertip, palm and nail recordings from a smartphone contain a signal that deep networks can use, and that, on this dataset, no fusion, pooling or demographic design choice has been shown to matter beyond the noise of the evaluation. It has also shown how much remains unresolved: the minority class is poorly detected, the binary labels used in part of the work must be corrected, one experiment must be repeated, and no external validation exists. These findings define a clear route from the present results to a method that could be evaluated for clinical use.

6.2 Limitation

The conclusions are limited in the following ways. The dataset comes from one site, one phone model and participants aged 18 to 25, and the reference hemoglobin was measured non-invasively, so generalization to other populations and devices is not supported. The test sets are small, and the Moderate class has only seven test participants, so most differences between configurations are within the range of chance variation. Only the best configuration of each unimodal ablation was reported, so its gains are optimistic estimates. Results from different groups of experiments are not directly comparable, because they differ in label space, test set and evaluation unit. The benchmarking uses three protocols, and only one can be compared with the selected model directly.

Two further problems affect the reliability of the results. First, the label file used for

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the binary single-modality experiments and for the binary fusion experiment is inconsistent with the participant metadata for 72 of 909 participants, which means that these results contain label noise, and the high accuracy of the binary fusion experiment in particular cannot be accepted until it is repeated with corrected labels. Second, the five-fold result of the BiLSTM configuration with demographic variables classified all Mild participants correctly, unlike every other configuration, and reads its images from different input folders; it was excluded from the conclusions and must be repeated on the common image folders. Third, the fusion comparison changes the head size together with the fusion rule, and the statistical tests available with five folds cannot reach conventional significance levels. The performance on the Moderate class and the sensitivity of the models are insufficient for stand-alone clinical screening. Explainability methods were considered in the planning of the project but were not pursued, so the models cannot at present show which image regions drive their predictions.

6.3 Future Work

The most urgent steps are to repeat the BiLSTM experiment with demographic variables on the common image folders and to compare the stored images class by class, and to repeat the binary experiments with labels taken from the participant metadata. After that, several directions follow from the findings. The selected model and the mean-pooling concatenation control should be re-evaluated with several random seeds and repeated cross-validation, with per-participant predictions saved so that paired bootstrap or McNemar tests become possible, and with the same head size for both fusion modules. The comparative benchmarking (Section 4.3.5) should be repeated with the same split for every method, several seeds or repeated cross-validation, and more Moderate participants before any ranking is claimed, and it should be extended with single-modality and leave-one-modality-out runs under the common protocol, parameter counts and training time. Shorter training or early stopping should also be tested, because the validation loss is lowest after the first epochs. Strategies aimed at the minority class, such as resampling, focal losses or a hierarchical decision between Normal and anemic classes, might increase the recall of Moderate anemia. The role of sex and of other confounders could be examined with stratified analyses. Larger datasets from several sites, devices and age groups, together with external validation against laboratory hemoglobin, are necessary before any statement on clinical usefulness. Finally, explainability analysis of the trained networks and a lightweight mobile implementation would be natural steps toward a screening application, provided that the sensitivity of the models is first improved.

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